# Superpixel-Based Reweighted Low-Rank and Total Variation Sparse Unmixing for Hyperspectral Remote Sensing Imagery

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***Abstract*— Sparse unmixing, as a semisupervised unmixing method, has attracted extensive attention. The process of sparse unmixing involves treating the mixed pixels of hyperspectral imagery as a linear combination of a small number of spectral signatures (endmembers) in a standard spectral library, associated with fractional abundances. Over the past ten years, to achieve a better performance, sparse unmixing algorithms have begun to focus on the spatial information of hyperspectral images. However, less accurate spatial information greatly limits the performance of the spatial-regularization-based sparse unmixing algorithms. In this article, to overcome this limitation and obtain more reliable spatial information, a novel sparse unmixing algorithm named superpixel-based reweighted low-rank and total variation (SUSRLR-TV) is proposed to enhance the performance of the traditional spatial-regularization-based sparse unmixing approaches. In the proposed approach, superpixel segmentation is adopted to consider both the spatial proximity and the spectral similarity. In addition, a low-rank constraint is enforced on the objective function as pixels within each superpixel have the same endmembers and similar abundance values, and they naturally satisfy the low-rank constraint. Differing from the traditional nuclear norm, a reweighted nuclear norm is used to achieve a more efficient and accurate low-rank constraint. Meanwhile, low-rank consideration is also used to enhance the spatial continuity and suppress the effects of random noise. Furthermore, TV regularization is introduced to promote the smoothness of the abundance maps. Experiments on three simulated data sets, as well as a well-known real hyperspectral imagery data set, confirm the superior performance of the proposed method in both the qualitative assessment and the quantitative evaluation, compared with the state-of-the-art sparse unmixing methods.**

***Index Terms*— Hyperspectral remote sensing, reweighted low-rank constraint, simple linear iterative clustering (SLIC), sparse unmixing, superpixels, total variation (TV).**

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## I. Introduction

With the rapid development of imaging technology, hyperspectral remote sensing, which involves collecting abundant spectral information with hundreds of contiguous bands, has been widely applied in various fields, such as mineral exploration, target detection, agricultural production, food safety, and military applications [1]–[5]. However, due to the limited spatial resolution of hyperspectral images (HSIs) and the complexity of the objects' distributions, mixed pixels are widely encountered in HSIs, which greatly affect further exploration of the information contained in HSIs. To solve the mixed pixel problem, the hyperspectral unmixing technique has been proposed to decompose the observed hyperspectral data into several different components of pure spectral signatures, named endmembers, and their corresponding proportions, named fractional abundances [6]–[9], which can better distinguish the different materials at the subpixel level.

Most hyperspectral unmixing algorithms consider the process of mixed pixel formation based on the typical linear mixture model (LMM) or a nonlinear mixture model (NLMM) [6], [10]. Some of the algorithms based on an NLMM [11]–[13] can consider the actual mixing process, such as the multiple reflections, spectral variability, and the intimate interactions of materials [14], [15]. However, most of these algorithms require *a priori* information of the actual mixing at the surface, which limits the practicability of these models [16]. The LMM, which assumes that the spectra collected by the sensor are an approximate result of a simple linear combination of endmembers weighted by their corresponding abundances, has aroused great interest from researchers [17]. Due to its simplicity, efficiency, and clear physical meaning, we focus on the LMM-based spectral unmixing algorithms in this article.

Based on the LMM, there are three typical unsupervised methods for hyperspectral unmixing that have been extensively studied [6], [18]: geometry-based methods [19]–[22], statistics-based algorithms [23], and nonnegative matrix factorization (NMF)-based approaches [24]–[26]. These unsupervised methods can directly extract the endmembers from the scene, without any prior knowledge. However, the assumption of pure pixels is often difficult to fulfill [19]–[21]. Accordingly, these algorithms may extract virtual endmembers that have no

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actual physical meaning from the highly mixed HSIs [27]. At the same time, the number of endmembers needs to be estimated before spectral unmixing in some of the above methods, which is also a challenging problem.

To address this issue, sparse regression has been proposed as a semisupervised spectral unmixing approach which can circumvent the endmember identification process [6], [28]. By introducing a spectral library containing potential pure spectral signatures as *a priori* knowledge, sparse regression can effectively avoid the negative impact of inaccurate endmember extraction on the performance of hyperspectral unmixing. For example, sparse unmixing by variable splitting and augmented Lagrangian (SUnSAL) algorithm [28] uses the $l_1$-norm to measure the sparsity of the abundance vector in each pixel, which is an approach that can achieve a superior unmixing performance, and the alternating direction method of multipliers (ADMM) [29] is also used to solve constrained sparse regression problems. Furthermore, the $l_p$-norm ($0 < p < 1$) and other nonconvex approximations have been shown to be closer to the $l_0$-norm (also in blind hyperspectral unmixing) [30], [31]. However, the pixel-by-pixel sparse unmixing methods consider the sparse structure of the abundance vector for each pixel in isolation, ignoring the fact that these pixels share a subset of the standard spectral library (containing the same endmembers). Through the observation that the $l_{2,1}$ mixed norm can accurately measure the row sparsity of the matrix, the collaborative SUnSAL (CSUnSAL) algorithm [32] uses the $l_{2,1}$-norm to ensure that all the abundance vectors exhibit global row sparsity. At the same time, by sharing the same sparse structure, this model is also robust to outliers. Proper use of the spatial prior information can effectively enhance the unmixing results when the spectral library correlation is high, or under conditions of high noise interference. Total variation (TV) [33] regularization takes advantage of the correlation of the first-order pixel neighborhood and can significantly improve the unmixing results. To obtain accurate spatial information, many algorithms focus on the spectral similarity of the HSIs rather than the simple proximity of the location. For example, Zhang *et al.* [34] proposed an unmixing model which contains only one regularization term, which is named spectral–spatial weighted sparse unmixing ($S^2$ WSU). In this algorithm, the spectral weighting factor uses an iterative reweighting approach to enhance the sparsity of the endmembers in the spectral library, and the spatial weighting factor uses the inverse of the sum of the local block of the abundance matrix to obtain similar abundance values in the block. Nevertheless, this method uses only a rough area to express the similarity and has an insufficient spatial penalty. When an HSI is under high noise interference conditions, the performance is poor. Based on robust sparse unmixing, the local spectral similarity preserving (LSSP) constraint was proposed by Li *et al.* [35], which is based on the idea that the higher the similarity of two pixels' spectra, the higher the similarity of the abundance values, thus obtaining the local similarity of the abundance map. However, as in SUnSAL-TV, LSSP considers only the correlation of the pixels in the first-order domain, and it ignores the relationships of other similar pixels. In recent years, the homogeneous region-based sparse unmixing methods have been intensely studied. For example, based on graph regularization, Li *et al.* [36] designed a superpixel construction model to enhance the similarity of each pixel within the superpixel block. However, because unmixing is performed on each superpixel separately, the resulting abundance maps are significantly different between each superpixel, even on some similar superpixels. To mitigate the computational complexity of large-scale HSIs and obtain spatial-contextual information, a fast multiscale spatial regularization unmixing algorithm (MUA) was proposed by Borsoi *et al.* [37]. In this algorithm, superpixel-based segmentation is performed before unmixing to form a coarse domain, and the unmixing result for the coarse domain forms a multiscale spatial regularization for unmixing of the original domain. All these algorithms use the spectral similarity and positional proximity (local similarity) of pixels to form the spatial prior information. The experimental results obtained in these studies have once again confirmed that the effective use of spatial information can significantly improve the unmixing performance.

The low-rank constraint [38] is a powerful tool that can be used to achieve the preservation of the low-dimensional structure of the image. As such, the low-rank constraint has been widely used in HSI processing, in applications such as HSI restoration [39], HSI super-resolution [40], HSI denoising [41], HSI anomaly detection [42], and HSI unmixing [43]–[47]. In [39]–[41] and [47], the low-rank constraint and TV regularization were combined in HSI processing, obtaining good results. In addition, Giampouras *et al.* [48] simultaneously enforced sparse and low-rank constraints on the entire abundance matrix and obtained the correlation of all the pixels. However, global abundance correlation rarely occurs in a scene, as endmembers are usually uniformly distributed in the homogeneous regions. For pixels in the homogeneous regions of an HSI, they contain not only the same endmembers but also similar abundance values, which make the abundance vectors of these pixels exhibit strong correlation. This fact also illustrates that the low-rank constraint is better enforced on local regions.

To address the above problem, a new sparse unmixing method named superpixel-based reweighted low-rank and total variation (SUSRLR-TV) is proposed in this article, in which superpixel segmentation is used to extract the homogeneous regions, and a low-rank constraint is imposed to promote the correlation of each superpixel's abundance matrix. At the same time, a nonisotropic TV model is adopted to impose the sparsity on the abundance in the domain of finite differences, which can improve the smoothness of the entire abundance map. To sum up, the proposed unmixing method makes two main contributions.

1) Block-based spatial–spectral information is considered in this work. Superpixel segmentation is first applied to obtain the homogeneous regions. A tighter reweighted low-rank constraint is then used to maintain the low-dimensional structure of each superpixel abundance matrix.
2) Sparsity, and first-order spatial correlation, is also enforced in the proposed method. A nonisotropic TV

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regularizer is used to apply sparsity on the difference domain of the abundance map, to increase the image smoothness. The two effective constraints are incorporated into the sparse unmixing framework and quickly solved by the ADMM.

To assess the effectiveness of the proposed method, four typical sparse unmixing methods, that is, SUnSAL, SUnSAL-TV, $S^2$WSU, and $\text{MUA}_{\text{SLIC}}$, were selected and compared with the proposed method, using three synthetic data sets with different noise levels, as well as a real HSI. The experimental results demonstrate that the proposed SUSRLR-TV algorithm can effectively overcome the loss of detail information, especially for pixels with small abundance values, and it can achieve a superior spectral unmixing accuracy.

The rest of this article is organized as follows. Section II presents the basic idea of sparse unmixing. The proposed SUSRLR-TV algorithm is described in detail in Section III. Section IV gives a description of the experimental data sets and analyzes the experimental results. Our conclusions are drawn in Section V.

## II. Sparse Unmixing

### A. Linear Mixture Model

The LMM assumes that the HSI can be represented as a linear combination of several pure spectral signatures, and it has been widely used in hyperspectral unmixing. Given an HSI with $n$ pixels, for one of the pixels, it can be written as follows:

$$\mathbf{y} = \mathbf{M}\mathbf{x} + \mathbf{n} \tag{1}$$

where $\mathbf{y} = [y_1, y_2, \ldots, y_l]^{\mathrm{T}}$ is a vector denoted as a spectral signature of a mixed pixel, and $l$ is the number of spectral bands. $\mathbf{M} = [\mathbf{m}_1, \mathbf{m}_2, \ldots, \mathbf{m}_p]$ is an $l \times p$ spectral signature matrix containing $p$ endmembers, and $\mathbf{m}_j$ is the $j$th endmember. $\mathbf{x} = [x_1, x_2, \ldots, x_p]^{\mathrm{T}}$ is a $p \times 1$ abundance vector, and $x_j$ represents the proportion of the $j$th endmember in this mixed pixel. $\mathbf{n} = [n_1, n_2, \ldots, n_l]^{\mathrm{T}}$ is an $l \times 1$ vector denoting the additional noise or the model error. Finally, two physical constraints are imposed—the abundance nonnegativity constraint (ANC) and the abundance sum-to-one constraint (ASC)—which make the LMM more practical. These two abundance constraints can be denoted as follows:

$$x_i \geq 0 \quad (i = 1, 2, \ldots, p) \tag{2}$$

$$\mathbf{1}^T\mathbf{x} = 1. \tag{3}$$

### B. Sparse Unmixing

When a standard spectral library is introduced as prior knowledge to circumvent the challenge of endmember identification, the traditional LMM can be extended to a sparse unmixing model. Because the endmember information has been previously obtained in the spectral library, the goal of the sparse unmixing is to find the best combination of observed spectra from the large spectral library. Equation (1) can then be rewritten as follows:

$$\mathbf{y} = \mathbf{A}\mathbf{x} + \mathbf{n} \tag{4}$$

where $\mathbf{A} \in R^{l \times m}$ is the standard spectral library containing the $m$ endmembers' signatures for unmixing. Because only a few endmembers in spectral library $\mathbf{A}$ participate in mixed spectra $\mathbf{y}$, $\mathbf{x}$ contains many zero values, which means that $\mathbf{x}$ is sparse. The original LMM can then be redefined as an $l_2$–$l_0$-norm optimization problem as follows:

$$\min_{\mathbf{x}} \|\mathbf{x}\|_0 \quad \text{s.t. } ||\mathbf{y} - \mathbf{A}\mathbf{x}||_2^2 \leq \delta, \quad \mathbf{x} \geq 0, \ \mathbf{1}^T\mathbf{x} = 1 \tag{5}$$

where $\delta$ represents the error of the reconstruction between $\mathbf{A}\mathbf{x}$ and mixed pixel $\mathbf{y}$. $\|\|_0$ is the $l_0$-norm, which denotes the number of nonzero components of a vector. However, the objective function (5) is difficult to solve directly as the $l_0$-norm is a typical NP-hard problem. Inspired by the theory of sparse representation, this problem can be solved by the convex relaxation strategy [49], as the $l_1$-norm is an effective alternative to the $l_0$-norm under a certain condition of the restricted isometric property (RIP) [50], [51]. Hence, this problem can be transformed into a standard convex optimization problem by means of the $l_1$-norm, as follows:

$$\min_{\mathbf{x}} \frac{1}{2}||\mathbf{y} - \mathbf{A}\mathbf{x}||_2^2 + \lambda\|\mathbf{x}\|_1 + l_{R^+}(\mathbf{x}) \tag{6}$$

where $||\mathbf{x}||_1 = \sum_{i=1}^{m} |x_i|$ represents the $l_1$-norm, and $x_i$ is the $i$th value of $\mathbf{x}$. $\lambda$ is a regularization parameter used to adjust the weight of the sparsity in this model. $l_{R^+}(\mathbf{x})$ is an indicator function that is infinite as long as the value of one of the abundance vectors $\mathbf{x}$ has a negative value; otherwise, the value of this function is zero, and this term is used as the ANC. It should be noted that the ANC is enforced explicitly, without the ASC, due to the fact that it has been pointed out in some studies that the ASC can have a negative impact [28], [33].

The early sparse unmixing methods paid too much attention to the spectral information and ignored the spatial information of the HSIs. Recently, more and more spatial regularization sparse unmixing algorithms have been proposed and have made great progress. In [33], TV was used as the correlation of the first-order domain of pixels to enhance the similarity of the local pixel abundance values, and the objective function of SUnSAL-TV was defined as follows:

$$\min_{\mathbf{X}} \frac{1}{2}||\mathbf{Y} - \mathbf{A}\mathbf{X}||_F^2 + \lambda||\mathbf{X}||_{1,1} + \lambda_{\text{TV}}\text{TV}(\mathbf{X}) + l_{R^+}(\mathbf{X}) \tag{7}$$

where $\mathbf{Y} = [\mathbf{y}_1, \mathbf{y}_2, \ldots, \mathbf{y}_n]$ denotes an HSI, and $n$ is the number of pixels. $\|\|_F^2$ is the Frobenius norm, and $||\mathbf{X}||_{2,1} = \sum_{i=1}^{m}\sum_{j=1}^{n} x_{i,j}^2$. Among them, $x_{ij}$ represents the abundance value in the $i$th row and the $j$th column in the abundance matrix $\mathbf{X}$.

$\mathbf{X} = [\mathbf{x}_1, \mathbf{x}_2, \ldots, \mathbf{x}_n]$ represents the abundance matrix, and $\|\|_F^2$ is the Frobenius norm. $\mathbf{x}_i$ represents the abundance vector of the $i$th pixel in the abundance matrix $\mathbf{X}$. $\text{TV}(\mathbf{X}) \equiv \sum_{i,j\in\delta} ||\mathbf{x}_i - \mathbf{x}_j||_1$, in which $\mathbf{x}_i$ and $\mathbf{x}_j$ are the abundance vectors of pixel $i$ and pixel $j$, respectively. $\delta$ is the set of (horizontal and vertical) pixel neighbors in $\mathbf{X}$. By minimizing the gradient sum of the horizontal and vertical directions of the abundance matrix, this method can not only consider the similarity of the abundance map but also achieve denoising for the randomly distributed noise. Although TV does preserve the boundaries well in some locations with relatively large

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abundance values, for over-large spatial parameters, TV does smooth some details [34]–[36].

**Algorithm 1** Simple Linear Iterative Clustering for Hyperspectral Imagery

**Input:** $\mathbf{Y}$, the number of superpixels $S_k$.
**Output:** Segmentation label $L$;
1. **Initialization:**
2. Each cluster center $\mathbf{C}_i = [y_i, x_i, y_i]^T$ is obtained at regular grid steps $S$;
3. Set $D'(j) = \infty$, $L(j) = -1$ for each pixel $j$;
4. Move each cluster center to the lowest spectral gradient position in a $3 \times 3$ neighborhood;
5. **Repeat:**
6. **for** each cluster center $\mathbf{C}_i$
7. **for**each pixel $j$ in the $2S \times 2$ $S$ range around $\mathbf{C}_i$
8. compute distance $dis\ (j)$ according to (8)
9. **if** $dis(j) < D'(j)$
10. $D'(j) = dis(j), \quad L(j) = i$
11. **end if**
12. **end for**
13. **end for**
14. **Update**new cluster center
15. **Until** the cluster center converges

## III. Sparse Unmixing via SUSRLR-TV

To obtain more accurate spatial information and avoid the appearance of an excessively smooth abundance map, both the positional relationship and the similarity of the spectra between pixels should be considered. Therefore, SUSRLR-TV is proposed using more accurate spatial information and an effective constraint strategy. Unlike the traditional methods that only consider the positional relationship between pixels (i.e., the first-order domain, the local window [33]–[35]), the SUSRLR-TV method can acquire far more accurate spatial information by the spectral similarity and spatial neighborhood from the original HSI, as adjacent pixels in an HSI tend to exhibit similar endmembers (except for pixels at the boundaries). Similar spectra will also be derived from the same endmembers with similar abundances.

The basic flowchart of the proposed method is shown in Fig. 1, where the well-known urban HSI [4] is taken as an example. In the first step, superpixels with semantic information are obtained by superpixel segmentation using the simple linear iterative clustering (SLIC) method [52]. The positional information of the pixels contained in these superpixels is then shared with the abundance matrix $\mathbf{X}$. To implement a more efficient low-rank constraint, two strategies are used. First, the abundance matrix of each superpixel is divided into several submatrices of the same size in columns. Second, unlike the traditional convex approximation, the reweighted nuclear norm is used to approximate the rank. At the same time, TV regularization is also used to enhance the smoothness of the abundance map and the correlation of similar superpixel groups. Furthermore, the nonisotropic TV can effectively promote the sparseness of the abundance map in the finite difference domain. Finally, the proposed objective function is efficiently solved by the ADMM strategy.

### A. Homogeneous Regions Based on Superpixel Segmentation

Pixels within a homogeneous region are spatially compact and tend to exhibit similar reflection spectra, indicating that these pixels may share the same endmembers and similar abundance values. Xu *et al.* [53] took note of this property of the homogeneous regions and divided the regions of the HSI. They then performed endmember extraction and abundance inversion separately in each region to obtain more accurate results. In the proposed approach, the similarity of pixels in a homogeneous region is further used by incorporating the strong linear correlation of the abundance vectors of these pixels in the sparse unmixing method, to obtain more local detail information. To obtain the homogeneous regions, superpixel segmentation can be introduced to form a series of regions containing unique semantic information (spatial information and spectral information). A superpixel is a group of adjacent pixels with similar color, texture, and other characteristics [54]. The SLIC method, which is fast and memory-efficient, can achieve state-of-the-art boundary adherence. Based on the SLIC method, the location information and color (or spectral) information can be better considered. In addition, very few preset parameters (only one parameter with a presegmented number of superpixels) need to be set, which is very practical. For the above reasons, in the proposed approach, the SLIC-based superpixel segmentation method is extended from RGB images to HSIs to acquire the homogeneous regions. At the same time, small-scale superpixels can be expected to form more accurate homogeneous regions.

The SLIC algorithm can be seen as a localized version of the $k$-means clustering method. For an HSI with $n$ pixels, if $S_k$ superpixels are desired, each superpixel will contain $n/S_k$ pixels. In the initial generation, the step size of the adjacent seed points is $S = \text{sqrt}(n/S_k)$, and the search window is fixed near the seed point of size $2\text{S} \times 2\text{S}$. To better distinguish the SLIC method from $k$-means clustering, Fig. 2 shows the difference, where the red points are the initial seed points. It can be found from Fig. 2 that during the clustering process, the SLIC algorithm only pays close attention to the pixels near the cluster center, whereas $k$-means considers the global pixels. As a result, SLIC has a lower time complexity. In the first generation of clustering, the SLIC algorithm fine-tunes the seed point to a gentle gradient nearby. After each generation of clustering is completed, cluster center vector $[y, x, y]^{\text{T}}$ ($x$, $y$ represent the position and $\mathbf{y}$ is the spectra) is updated by averaging the spectral and positional information of all the clustered pixels in the current region. When the cluster center does not change significantly, the iterative process of the SLIC algorithm stops. Differing from the strategy applied in RGB images, the luminosity and color component of the pixels in CIELAB space (the transformed space used in the SLIC method) are replaced by the spectral vector, to extend the method to HSIs. By considering both the spectral distance and the spatial distance, the current pixel is clustered to the nearest

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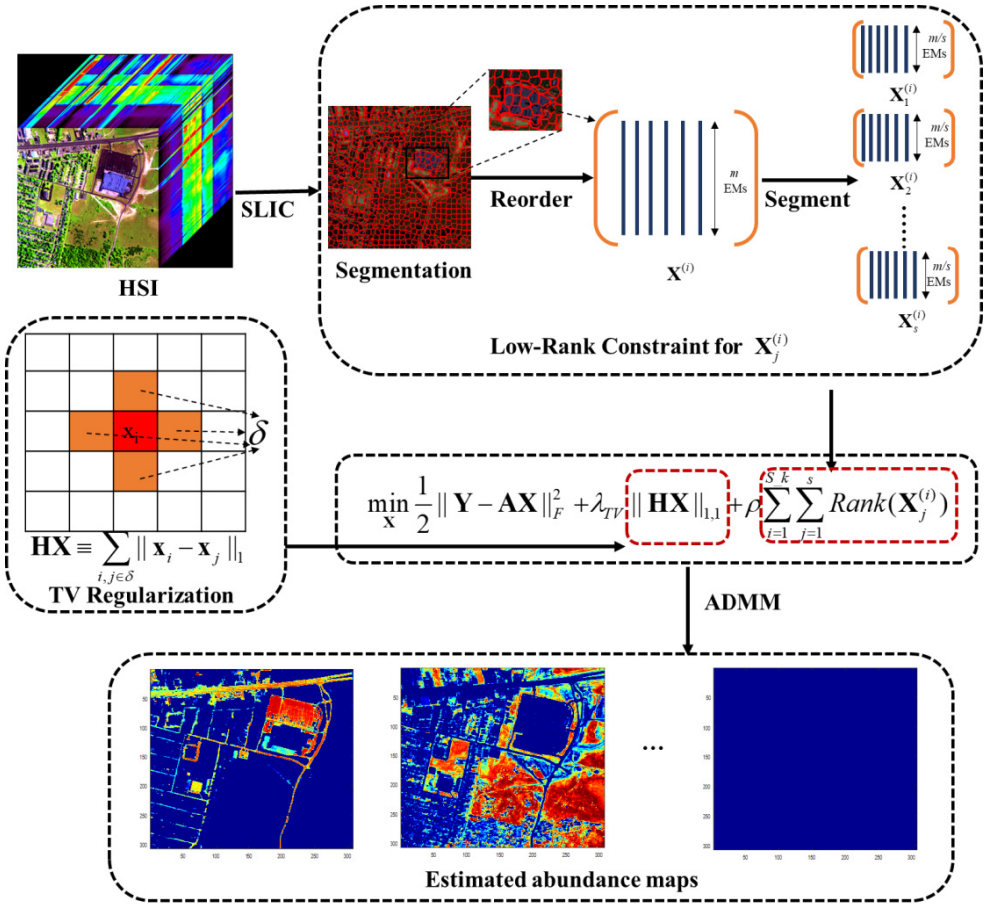



Fig. 1. Basic flowchart of sparse unmixing through SUSRLR-TV, taking urban HSI as an example.

seed point. These distance metrics are defined as follows:

$$d_{\text{spatial}}(i,j) = \sqrt{(x_i - x_j)^2 + (y_i - y_j)^2}$$
$$d_{\text{spectral}}(i,j) = ||\mathbf{y}_i - \mathbf{y}_j||_2^2$$
$$D' = \sqrt{\left(\frac{d_{\text{spatial}}}{S}\right)^2 + \left(\frac{d_{\text{spectral}}}{m}\right)^2} \quad (8)$$

where $(x_i, y_i)$, $(x_j, y_j)$ represent the position, $\mathbf{y}_i$, $\mathbf{y}_j$ are the spectra of pixels $i$ and $j$, respectively, and $m$ is a balancing weight. The specific steps of the SLIC method for hyperspectral imagery are shown in Algorithm 1.

The selection of $S_k$ is very important, as an over-large value may generate small-size superpixels containing few pixels, and few pixels are not conducive to the use of the abundance vector correlation. If $S_k$ is set as too small a value, some unrelated pixels are likely to be classified into the same superpixel. Using these poorly correlated abundance vectors may lead to poor unmixing effects.

### *B. Reweighted Low-Rank Constraint*

Based on the fact that only a small subset of the standard spectral library is involved in the hyperspectral unmixing problem, the sparsity constraint has been found useful for imposing on abundance vectors and can obtain more precise abundance images. However, the low-dimensional structure of the image is not well-maintained in the traditional sparse unmixing approach, and the contribution of the low-rank property attribute is ignored. In a homogeneous region, the abundance vectors of these pixels not only share the same endmembers but also have very similar abundances. By applying the low-rank constraint to the abundance matrix formed by each homogeneous region, the local low-dimensional structure can be better maintained, which means more detailed features of the abundance map can be obtained. To enhance the low-rank properties of the active endmembers, each homogeneous

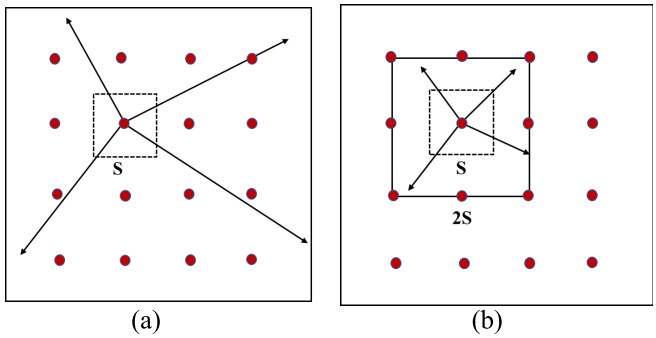



Fig. 2. Difference between SLIC and $k$-means clustering. (a) $k$-means clustering searches the entire image. (b) SLIC searches the surrounding area of 2S × 2S.

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region's abundance matrix is divided into several submatrices by columns. The abundances of the active endmembers then become dominant in each submatrix. This helps strengthen the correlation of the active endmembers in each homogeneous region.

By just considering this constraint, the unmixing model can be formed as follows:

$$\min_{\mathbf{X}} \frac{1}{2}||\mathbf{Y}-\mathbf{AX}||_F^2+\rho\sum_{i=1}^{S_k}\sum_{j=1}^{s}\mathrm{Rank}\left(\mathbf{X}_j^{(k)}\right). \tag{9}$$

$\mathbf{X}=[\mathbf{X}^{(1)},\ \mathbf{X}^{(2)},\ldots,\mathbf{X}^{(S_k)}]$, where $S_k$ is the number of superpixels and $\mathbf{X}^{(i)}$ represents the abundance matrix formed by all the pixels in the $i$th superpixel. The matrix $\mathbf{X}^{(i)}$ is divided into several submatrices by column, and $\mathbf{X}^{(i)}=[\mathbf{X}_1^{(i)};\mathbf{X}_2^{(i)};\ldots;\mathbf{X}_s^{(i)}]$, in which $\mathbf{X}_j^{(i)}$ is the $j$th submatrix in the $i$th superpixel, and $s$ is the number of submatrices in each superpixel.

It should be noted that the number of nonzero singular values of the matrix is equal to the rank of the matrix [55]. Assuming that $S(\mathbf{X})=[\sigma_1(\mathbf{X}),\sigma_2(\mathbf{X}),\ldots,\sigma_j(\mathbf{X}),\ldots]$ represents a singular value vector of matrix $\mathbf{X}$, where $\sigma_j(\mathbf{X})$ represents the $j$th singular value, then the original objective function can be rewritten as follows:

$$\min_{\mathbf{X}} \frac{1}{2}||\mathbf{Y}-\mathbf{AX}||_F^2+\rho\sum_{i=1}^{S_k}\sum_{j=1}^{s}\left\|S\left(\mathbf{X}_j^{(i)}\right)\right\|_0. \tag{10}$$

This is also an $l_F-l_0$ optimization problem, which is a difficult combinatorial optimization problem. Inspired by sparse representation theory [49], a nuclear norm, denoted as $\|\|_*$(sum of the singular values), is used to replace the nonconvex and discontinuous $l_0$-norm for singular values. Based on this idea, the problem is then transformed into a convex and derivable function.

To overcome the imperfect measurement of the traditional $l_1$-norm, the sparsity can be further enhanced using the reweighted $l_1$-norm [56]. The same processing can also be applied to the singular values of the matrix, to measure the sparsity of the singular value vectors [57]. $||\mathbf{X}||_{w,*}=\sum_{i=1}^{\mathrm{Rank}(\mathbf{X})}w_i\sigma_i(\mathbf{X})$ denotes a weighted nuclear norm, where $w_i$ is a weighting factor. $\sigma_i(\mathbf{X})$ is the $i$th singular value of matrix $\mathbf{X}$. In the iterative process of the algorithm, such as the $(k+1)$th iteration, $w_i=(1/\sigma_i^k(\mathbf{X})+\varepsilon)$ and $\sigma_i^k(\mathbf{X})$ represent the singular value of $\mathbf{X}$ at the $k$th iteration. $\varepsilon$ is a small positive number. If the singular value is close to zero, it will tend to obtain a smaller value with the penalty of $w_i$ in the next iteration. In the initial generation, this weighting factor is set to $\mathbf{w}=[1,1,\ldots,1]^{\mathrm{T}}$, which is a standard nuclear norm. Problem (10) is then converted into the following form:

$$\min_{\mathbf{X}} \frac{1}{2}||\mathbf{Y}-\mathbf{AX}||_F^2+\rho\sum_{i=1}^{S_k}\sum_{j=1}^{s}||\mathbf{X}_j^{(i)}||_{w,*}. \tag{11}$$

The application of the low-rank constraint to each superpixel can promote the correlation of pixels within the superpixel. However, the weak correlation of pixels between superpixels is also important [58]. To further enhance the smoothing effect of the abundance map and consider the connection of the abundance values between each superpixel, TV regularization is also imposed in the proposed model. Based on the above analysis, the final objective function is constructed as follows:

$$\min_{\mathbf{X}} \frac{1}{2}||\mathbf{Y}-\mathbf{AX}||_F^2+\lambda_{\mathrm{TV}}||\mathrm{TV}(\mathbf{X})||_{1,1}+\rho\sum_{i=1}^{S_k}\sum_{j=1}^{s}||\mathbf{X}_j^{(i)}||_{w,*}+l_{R^+}(\mathbf{X}). \tag{12}$$

It can be observed that the proposed model does not involve the traditional $l_1$-norm, or other constraints close to the $l_0$-norm, to guarantee the sparsity of the abundance matrix. The main reason for this is that the adopted TV regularization is nonisotropic TV based on the $l_1$-norm, which can effectively promote the sparsity of the abundance matrix in the finite difference domain. The sparsity of the abundance matrix in the finite difference domain produces two scenarios. The first is that most of the coefficients of the abundance of the endmembers are zero. The second is that most of the coefficients are nonzero. For nonzero situations, this is acceptable for the active endmembers. For inactive endmembers, most of the coefficients will not be zero, which leads to an increase in the residuals of the data fitting terms. At this time, the model tends to accept the case where the coefficients are zero. As a result, the nonisotropic TV can indirectly promote the sparsity of the abundance matrix.

Finally, the ADMM strategy is used to solve the optimization function (12). The specific steps of the ADMM are shown in Algorithm 2. More details can be found in the Appendix.

From Algorithm 2, it can be seen that the computational complexity of this algorithm depends on the update of variable $\mathbf{U}$. Because $\mathbf{A}^T\mathbf{A}+3\mathbf{I}$ is fixed, the complexity of the entire algorithm depends on the update of each auxiliary variable. Clearly, the singular value decomposition (SVD) operator consumes the most time. The complexity of SVD is $\mathcal{O}(m^2KS_k)$ at each iteration. Here, we recall that $K$ is the number of pixels in each superpixel, and $m$, $S_k$ are the number of endmembers and superpixels, respectively. Because $n=S_k\times K$, where $n$ is the number of pixels, the complexity can be written as $\mathcal{O}(m^2n)$. $\mathbf{V}_3$ has the order $\mathcal{O}(ln\log n)$, where $l$ is the number of bands, while the complexity of updating the other variables is $\mathcal{O}(n)$. Therefore, the computational complexity of this algorithm is $\mathcal{O}(m^2n)$.

## IV. Experiments and Analysis

In the experiments, three simulated data sets with different characteristics and a widely used real hyperspectral remote sensing image were used to evaluate the performance of the proposed SUSRLR-TV algorithm, compared with four state-of-the-art sparse unmixing algorithms, that is, SUnSAL [28], SUnSAL-TV [33], $S^2$WSU [34], and $\mathrm{MUA_{SLIC}}$ [37]. What needs to be specially stated is that the last three algorithms all consider spatial information to obtain a better unmixing performance, and the last one also uses SLIC to obtain superpixels, to consider the spatial-contextual information.

The signal-to-reconstruction-error (SRE) is used to quantitatively evaluate the performance of each algorithm. The

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**Algorithm 2** ADMM Pseudocode for SUSRLR-TV

**Input:** $\mathbf{Y}$, $\mathbf{A}$, the number of superpixels $S_k$ ;
**Perform**: the SLIC algorithm on $\mathbf{Y}$ to obtain superpixel groups $[\mathbf{X}^{(1)}, \mathbf{X}^{(2)}, \ldots, \mathbf{X}^{(S_k)}]$
**Output:** The estimated abundance matrix $\hat{\mathbf{X}}$;
**Initialization:** set $k = 0$, *Max_Iter*, choose $\mu > 0$, $\mathbf{U}^{(0)}, \mathbf{V}_1^{(0)}, \ldots, \mathbf{V}_5^{(0)}, \mathbf{D}_1^{(0)}, \ldots, \mathbf{D}_5^{(0)}$;
**Repeat:**
$\mathbf{U}^{(k+1)} \leftarrow \arg\min_{\mathbf{U}} L(\mathbf{U}, \mathbf{V}_1^{(k)}, \mathbf{V}_2^{(k)}, \mathbf{V}_3^{(k)}, \mathbf{V}_4^{(k)}, \mathbf{V}_5^{(k)}, \mathbf{D}^{(k)})$
  **for** $i = 1$: 5 && $i \neq 4$
$\mathbf{V}_i^{(k+1)} \leftarrow \arg\min_{\mathbf{V}_i} L(\mathbf{U}^{(k+1)}, \mathbf{V}_1^{(k)}, \ldots, \mathbf{V}_i, \ldots, \mathbf{V}_5^{(k)}, \mathbf{D}^{(k)})$
  **end for**
  **for** $i = 1$: $S_k$
  **for** $j = 1$: $s$
$(\mathbf{V}_4^{(k+1)})_j^{(i)} \leftarrow \arg\min_{(\mathbf{V}_4)_j^{(i)}} \rho||(\mathbf{V}_4)_j^{(i)}||_{w,*} + \frac{\mu}{2}||(\mathbf{U}^{(k)} - \mathbf{V}_4 - \mathbf{D}_4^{(k)})_j^{(i)}||_F^2$
  **end for**
  **end for**
  **Update Lagrange multipliers:** $\mathbf{D}^{(k+1)} \leftarrow \mathbf{D}^{(k)} - \mathbf{G}\mathbf{U}^{(k+1)} - \mathbf{B}\mathbf{V}^{(k+1)}$
  **Update iteration** $k \leftarrow k + 1$
**Until** some stopping criterion is satisfied
**Return** $\hat{\mathbf{X}} = \mathbf{U}^{(k)}$

definition of SRE is as follows:

$$\mathrm{SRE} \equiv 10\log_{10}\left(E\left[||\mathbf{x}||_2^2\right] / E\left[||\mathbf{x} - \hat{\mathbf{x}}||_2^2\right]\right) \tag{13}$$

where $\mathbf{x}$ represents the true abundance vector of the endmember, and $\hat{\mathbf{x}}$ is the estimated abundance vector by different algorithms. The better the performance of the algorithm, the higher the SRE value.

### *A. Experiments on Simulated Data*

The spectral library $\mathbf{A} \in \mathrm{R}^{224\times240}$ used in our experiments consisted of 240 spectral signatures with 224 bands, evenly distributed in the wavelength range of 0.4–2.5 $\mu$m, which were selected from the U.S. Geological Survey (USGS) spectral library. $p$ endmembers were picked from this spectral library $\mathbf{A}$ for generating the simulated data sets. In addition, the simulated data sets followed the LMM and were enforced with the ANC and ASC. Finally, white Gaussian noise was applied using a normalized cutoff frequency of $5\pi/l$, with different intensities of signal-to-noise ratio (SNR) for each set of data.

*1) Simulated Data Cube 1 (DC1):* DC1, which was made up of 75 × 75 pixels and 224 bands per pixel, was formed by selecting five endmembers from spectral library $\mathbf{A}$. As shown in Fig. 3, the abundance values of the background pixels for the five spectral signatures were fixed to [0.1149, 0.0741, 0.2004, 0.2055, 0.4051]. From top to bottom, for the square area, the fractional abundances were set as 1, a mixture of two endmembers (1/2 and 1/2, respectively), a mixture of three endmembers (1/3, 1/3, and 1/3, respectively), a mixture of four endmembers (1/4, 1/4, 1/4, and 1/4, respectively), and a mixture of five endmembers (1/5, 1/5, 1/5, 1/5, and 1/5, respectively). The spectral curves of the five endmembers are shown in Fig. 3(a).

*2) Simulated Data Cube 2 (DC2):* DC2 was generated using the same pattern as DC1, except that 15 endmembers were selected from the spectral library $\mathbf{A}$. Moreover, some endmembers had very small abundance values in this simulated data cube. More details are shown in Fig. 4. In addition, DC2 has previously been experimented with in [59] and can well reflect the advantages and disadvantages of the different sparse unmixing algorithms.

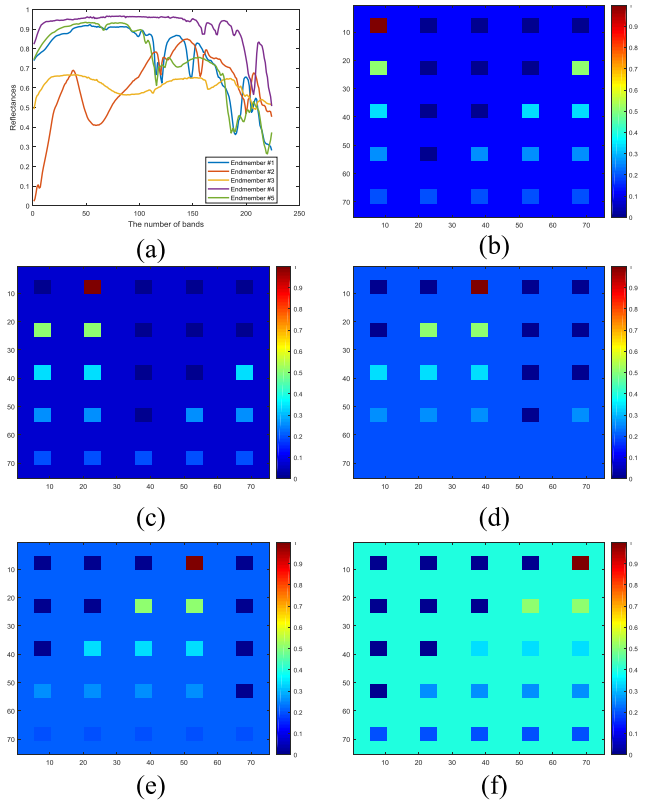



Fig. 3. Abundance maps of simulated DC1. (a) Spectra of the five endmembers. (b) Endmember 1. (c) Endmember 2. (d) Endmember 3. (e) Endmember 4. (f) Endmember 5.

*3) Simulated Data Cube 3 (DC3):* DC3 was made up of 100 × 100 pixels with 224 bands per pixel, with nine endmembers

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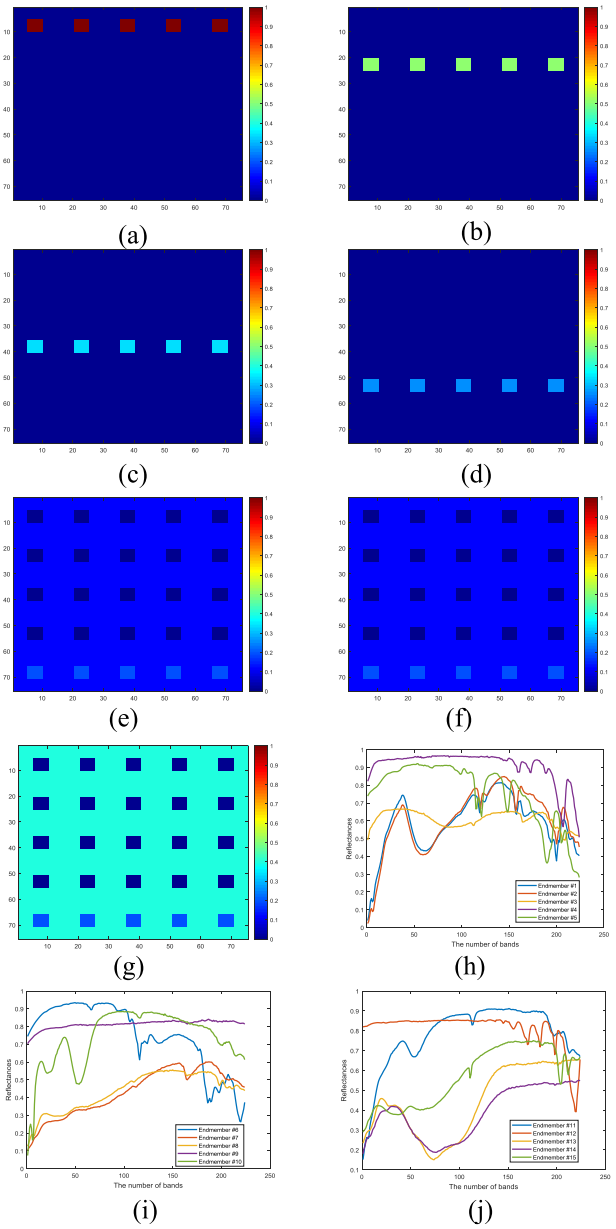



Fig. 4. Abundance maps of simulated DC2. (a) Endmember 1. (b) Endmember 2. (c) Endmember 4. (d) Endmember 7. (e) Endmember 11. (f) Endmember 13. (g) Endmember 15. (h) Spectra of endmembers #1–#5. (i) Spectra of endmembers #6–#10. (j) Spectra of endmembers #11–#15.

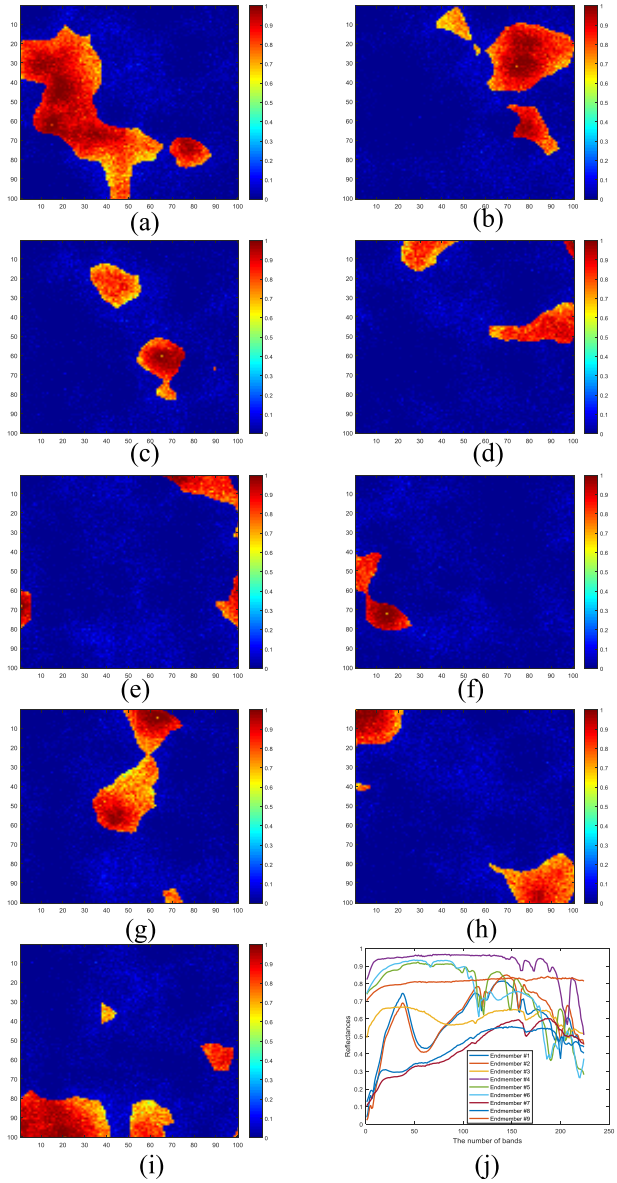



Fig. 5. Abundance maps of simulated DC3. (a) Endmember 1. (b) Endmember 2. (c) Endmember 3. (d) Endmember 4. (e) Endmember 5. (f) Endmember 6. (g) Endmember 7. (h) Endmember 8. (i) Endmember 9. (j) Spectra of endmembers #1–#9.

being selected from spectral library **A**. Unlike DC1 and DC2, the fractional abundances of DC3 were generated according to a Dirichlet distribution centered at a Gaussian random field [37]. This data set had a good spatial homogeneity area and sharp transitions, which could reveal the spatial features quite well for different unmixing algorithms. Further details of the fractional abundances are provided in Fig. 5.

The proposed SUSRLR-TV algorithm and the four other algorithms were jointly tested on the three simulated data sets. To ensure the fairness of the experiments, the proposed algorithm used the same segmentation strategy as $\text{MUA}_{\text{SLIC}}$ to build the superpixel structure. In addition, for DC1 and DC3, SUnSAL, SUnSAL-TV, $\text{S}^2$WSU, and $\text{MUA}_{\text{SLIC}}$ followed the best parameter settings in [37]. With regard to the parameter settings for DC2, different combinations of parameter settings were tested for each algorithm, to ensure the best performance. The specific parameter settings used in the simulated experiments are listed in Table I.

Fig. 6 shows the segmentation results for the three sets of simulated data using the SLIC method, with the SNR equal to 20, 30, and 20 dB, respectively. Fig. 6(a), (c), and (e) are false-color images composed of the three bands in the original HSIs. It can be seen that Fig. 6(a) loses a lot of detail at 20 dB, and the boundary of the square area is blurred. DC2 (SNR = 30 dB) shows a relatively high SNR compared with DC1 (SNR = 20 dB), and DC3 (SNR = 20 dB) is a less mixed data set, which is beneficial to obtaining accurate segmentation results, as shown in Fig. 6(d) and (f). Fig. 6(b)

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TABLE I
SPECIFIC PARAMETER SETTINGS USED IN THE SIMULATED EXPERIMENTS

| Algorithm | Parameters | | | | | | | | | |
|---|---|---|---|---|---|---|---|---|---|---|
| | Data / Pa | DC1 (SNR) | | | DC2 (SNR) | | | DC3 (SNR) | | |
| | | 10 | 20 | 30 | 20 | 30 | 40 | 10 | 20 | 30 |
| SUnSAL[28] | $\lambda$ | 1.5 | 7e−1 | 1e−1 | 5e−2 | 5e−3 | 5e−5 | 5e−1 | 7e−2 | 1e−2 |
| SUnSAL-TV[33] | $\lambda$ | 7e−2 | 5e−2 | 7e−3 | 7e−2 | 1e−4 | 5e−5 | 1e−1 | 1e−2 | 5e−3 |
| | $\lambda_{tv}$ | 1e−1 | 5e−2 | 1e−2 | 3e−1 | 5e−3 | 1e−3 | 5e−1 | 3e−2 | 7e−3 |
| $S^2$WSU[34] | $\lambda$ | 1 | 1e−1 | 5e−3 | 1 | 1e−2 | 1e−2 | 5e−2 | 1e−3 | 1e−3 |
| $MUA_{SLIC}$[37] | $\lambda_{sp1}$ | 5e−2 | 3e−2 | 7e−3 | 1e−2 | 1e−2 | 5e−4 | 1e−2 | 7e-3 | 3e-3 |
| | $\lambda_{sp2}$ | 5e−1 | 1e−1 | 5e−2 | 5e−1 | 1e−4 | 5e−3 | 5e−1 | 1e−1 | 3e−2 |
| | $\beta$ | 40 | 30 | 10 | 40 | 30 | 10 | 15 | 10 | 3 |
| | $S_k$ | 225 | 225 | 225 | 225 | 225 | 225 | 170 | 170 | 170 |
| SUSRLR-TV | $\lambda_{tv}$ | 1e−1 | 5e−2 | 1e−1 | 5e−1 | 5e−2 | 1e−3 | 1e−1 | 3e−2 | 7e−3 |
| | $\rho$ | 1e−1 | 5e−2 | 1e−2 | 5e−2 | 5e−3 | 1e−3 | 5e−1 | 5e−2 | 1e−2 |
| | $S_k$ | 225 | 225 | 225 | 225 | 225 | 225 | 170 | 170 | 170 |

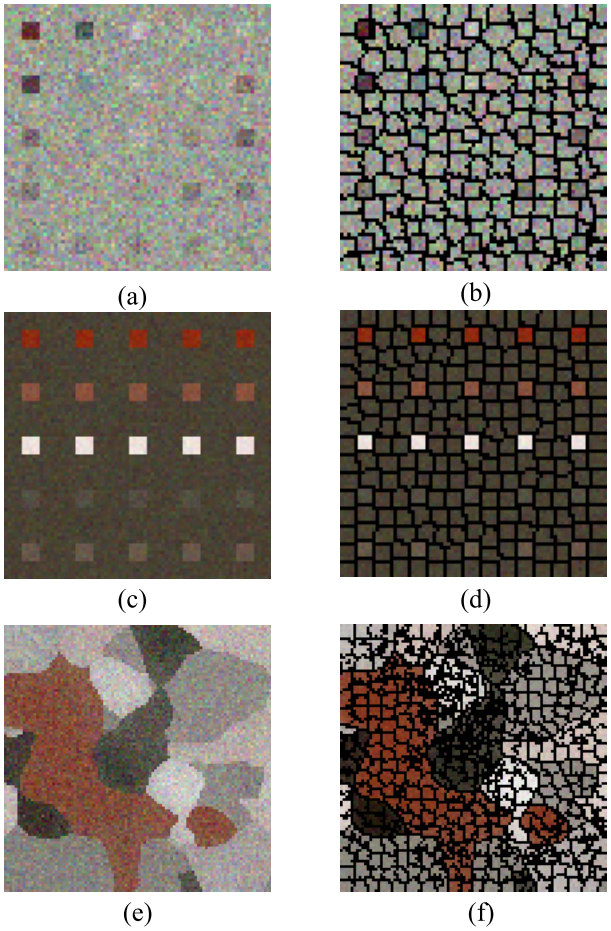



Fig. 6. Results of segmentation using the SLIC method on the simulated data. (a) False-color image of DC1 (SNR = 20 dB). (b) Segmentation (the number of superpixels is 224). (c) False-color image of DC2 (SNR = 30 dB). (d) Segmentation (the number of superpixels is 225). (e) False-color image of DC3 (SNR =30 dB). (f) Segmentation (the number of superpixels is 170).

still shows a relatively satisfactory result, and most of the square areas can be selected completely. However, the shape of these square areas shows slight distortion. It can be observed that the SLIC-based superpixel segmentation algorithm can suppress the noise pollution and obtain more accurate spatial information, for the final unmixing.

For a visual comparison, Figs. 7–9 show the estimated abundance maps for some of the endmembers for the five algorithms on the three simulated data sets. The three data sets were simulated with appropriate SNR, that is, 20, 30, and 20 dB, respectively. Among the three data sets, DC2 was set with a higher SNR because there were more active endmembers in this data set.

Figs. 7 and 8 show three representative estimated abundance maps. As no spatial information is considered in SUnSAL, it obtains the worst solutions on both data sets. Many isolated noise points are found in the abundance map of EM#5 in Fig. 7, and also in the abundance map of EM#12 in Fig. 8, confirming the drawbacks of the pixel-by-pixel unmixing method. In addition, the $S^2$WSU algorithm also exhibits some noise points in the abundance maps because it considers only the abundance relationship in a local window and enforces an insufficient spatial penalty. SUnSAL-TV, which considers the spatial consistency, achieves better results, in which the boundaries of some of the square areas can be clearly distinguished (EM#5 in Fig. 7). However, due to the unreliable spatial correlation, significant over-smoothing occurs in some of the estimated abundance maps. Furthermore, many of the square areas are mixed with the background and are difficult to distinguish (EM#1 and EM#2 in Fig. 7; EM#7 and EM#12 in Fig. 8). With different ways of considering the spatial information, SUSRLR-TV and $MUA_{SLIC}$ show more detailed information in the estimated abundance maps, based on the similarity of the homogeneous regions. In particular, more clear square regions appear in the estimated abundance maps of EM#1 and EM#2 in Fig. 7, together with EM#7 in Fig. 8, which are mostly blurred or even disappear with the previous three unmixing algorithms. However, compared with $MUA_{SLIC}$, quality improvements in some of the abundance maps (EM#1 and EM#5 in Fig. 7; EM#12 in Fig. 8) can still be clearly found with the proposed algorithm.

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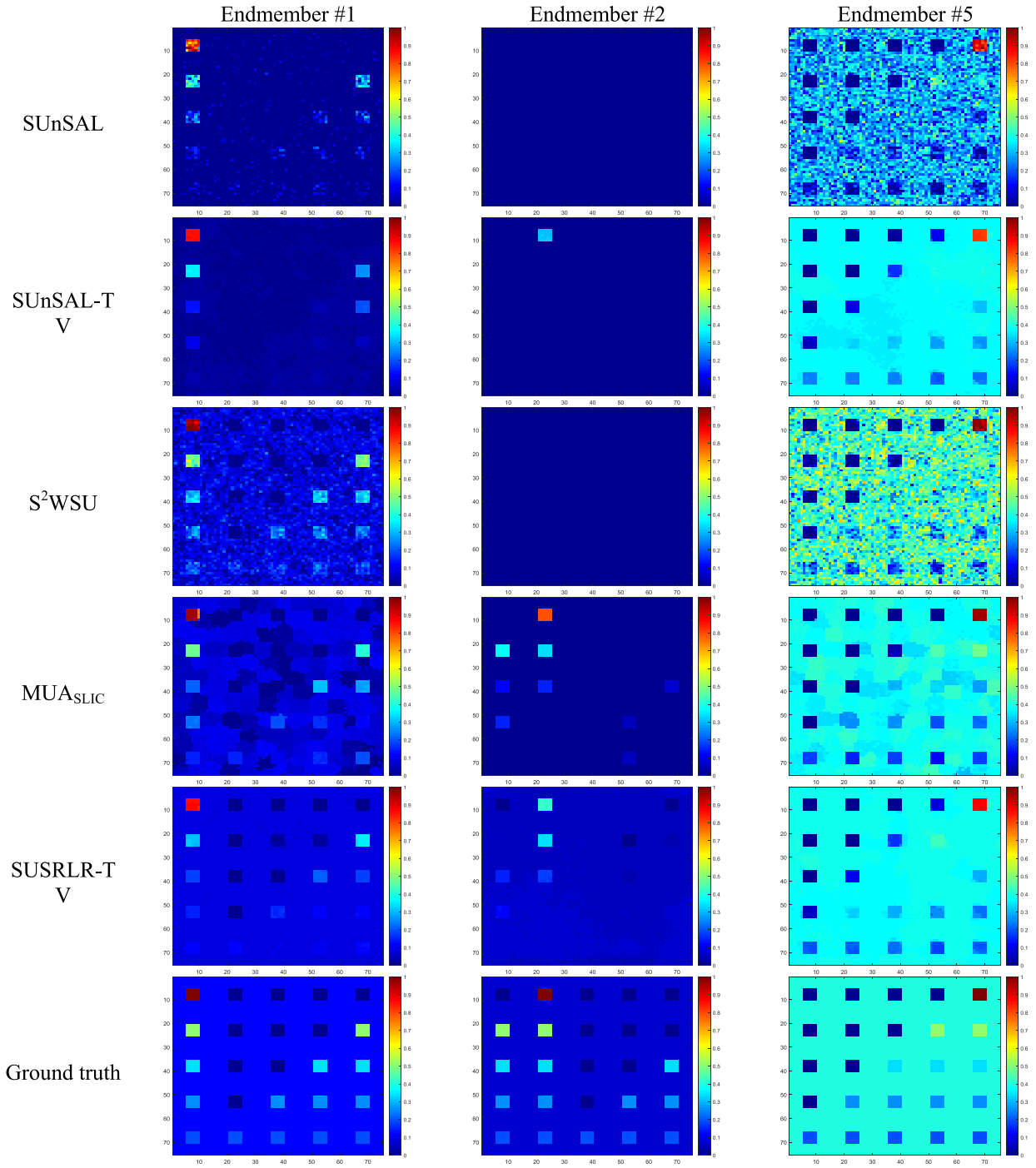



Fig. 7. Estimated abundance maps for endmembers #1, #2, and #5 under 20 dB of white noise on DC1.

The characteristic of a blocky distribution also appears in the abundance maps obtained by the $\text{MUA}_{\text{SLIC}}$ algorithm. Using superpixel segmentation to explore reliable spatial information, and by reweighting the low-rank constraint to achieve a close approximation of the local correlation of the abundance matrix, more accurate abundance maps are achieved by the proposed SUSRLR-TV algorithm. Furthermore, by adding the constraint of the TV regularization, the smoothness of the estimated abundance maps is further improved.

The results for DC3 are analyzed separately. The estimated abundance maps of each algorithm for EM#1, EM#4, and EM#9 in DC3 are shown in Fig. 9. Compared with the SUnSAL and $\text{S}^2$WSU algorithms, the other three algorithms generate abundance maps with fewer outliers. The abundance maps obtained by SUnSAL-TV exhibit obvious boundaries in the different homogeneous regions, but the abundance maps change too slowly with the abundance values of the three endmembers. SUnSAL-TV yields some distortion in transition regions and structural details (especially for EM#1). The

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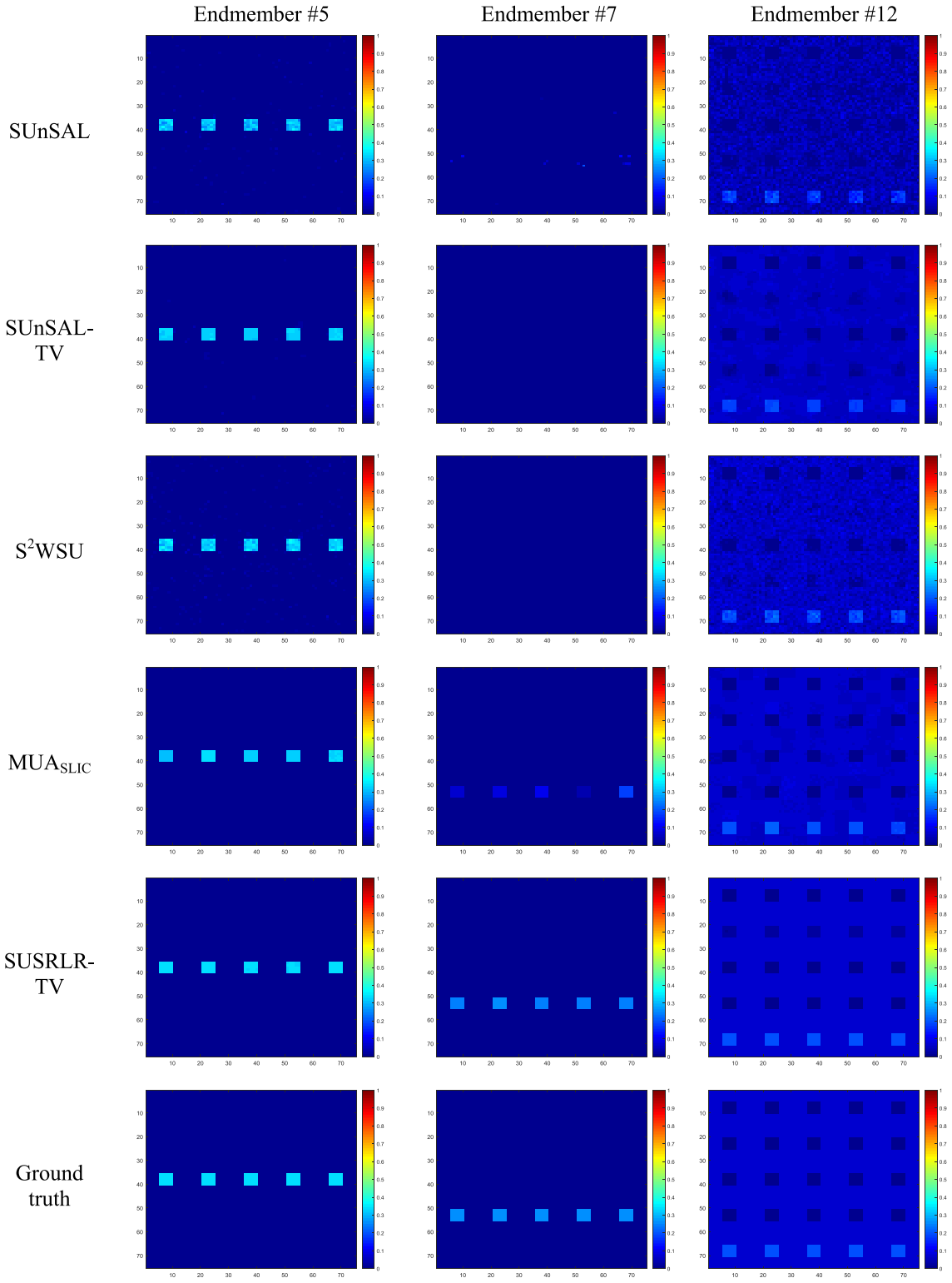



Fig. 8. Estimated abundance maps for endmembers #5, #7, and #12 under 30 dB of white noise on DC2.

$\text{MUA}_{\text{SLIC}}$ algorithm can achieve a mutation of the abundance values relative to the SUnSAL-TV algorithm, but the contour and shape are not as good as for the proposed algorithm (i.e., EM#1 and EM#9). In general, as SUnSAL does not consider the spatial information, it achieves the worst results on the simulated data. It can be clearly seen from these results that the use of spatial information can significantly improve the quality of the abundance maps. Overall, the proposed algorithm obtains abundance maps that are the closest to the ground truth on all three sets of simulated data.

Table II lists the SRE values of each algorithm at different SNRs on the simulated data sets. It can be found that at

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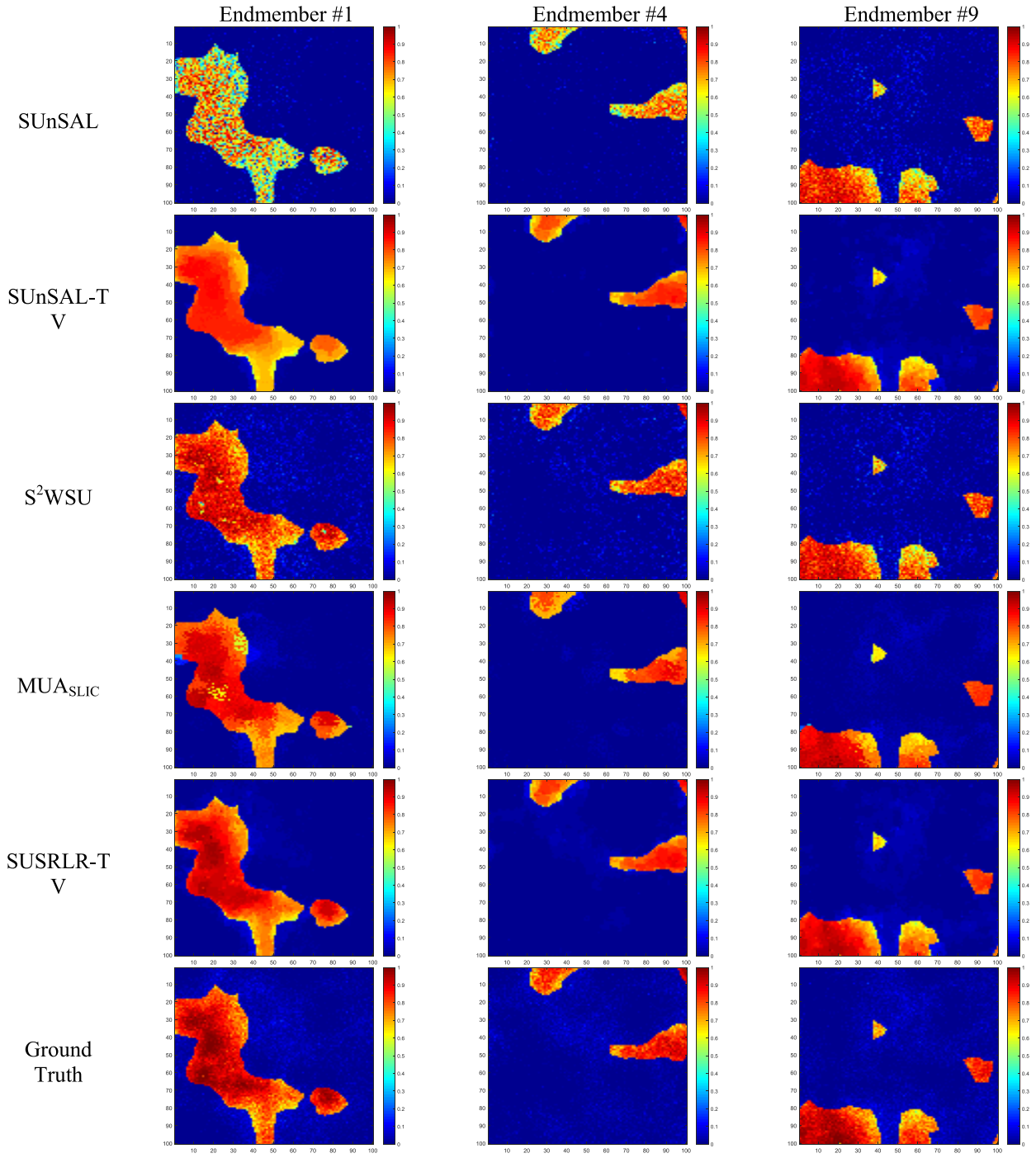



Fig. 9. Estimated abundance maps of endmembers #1, #4, and #9 under 20 dB of white noise on DC3.

higher SNR levels, all the algorithms obtain larger SRE values. All the other methods obtain far better quantitative results than SUnSAL, which indicates that the sparse unmixing methods considering the spatial information can achieve a satisfactory unmixing effect, in most cases. However, when the noise level is further increased, the difference between various algorithms is amplified. Among the different algorithms, the robustness of S2WSU is the worst among the algorithms considering spatial–spectral information. According to Table II, all three unmixing methods that consider spatial information show good antinoise capabilities and achieve higher SRE values. The proposed SUSRLR-TV method obtains the highest SRE values, except for DC3 at 10 dB. From the analysis of the simulated data sets, the proposed algorithm achieves the best results, both qualitatively and quantitatively.

### *B. Real Hyperspectral Data Experiment*

To further verify the performance of the proposed algorithm, it was necessary to conduct an experiment on a real HSI. The well-known Airborne Visible Infrared Imaging Spectrometer

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TABLE II
SRE VALUES ACHIEVED BY DIFFERENT UNMIXING METHODS ON THE SIMULATED DATA

| Method / SNR | SUnSAL | SUnSAL-TV | $S^2$WSU | $MUA_{SLIC}$ | SUSRLR-TV |
|---|---|---|---|---|---|
| **DC1 data cube** | | | | | |
| **10 dB** | 0.05 dB | 4.88 dB | 0.29 dB | 6.21 dB | **7.59** dB |
| **20 dB** | 4.54 dB | 9.42 dB | 7.69 dB | 11.17 dB | **14.33** dB |
| **30 dB** | 8.90 dB | 14.44 dB | 15.49 dB | 15.73 dB | **24.98** dB |
| **DC2 data cube** | | | | | |
| **20 dB** | 0.84 dB | 2.79 dB | 1.92 dB | 4.62 dB | **10.81** dB |
| **30 dB** | 2.69 dB | 8.11 dB | 5.04 dB | 9.80 dB | **24.58** dB |
| **40 dB** | 3.92 dB | 13.44 dB | 16.60 dB | 16.62 dB | **35.68** dB |
| **DC3 data cube** | | | | | |
| **10 dB** | 1.33 dB | 2.13 dB | 3.24 dB | **7.16** dB | 5.01 dB |
| **20 dB** | 4.28 dB | 11.55 dB | 9.33 dB | 12.41 dB | **15.74** dB |
| **30 dB** | 10.44 dB | 17.99 dB | 21.67 dB | 18.33 dB | **22.27** dB |

(AVIRIS) Cuprite data set [33] was adopted to verify the effectiveness of the proposed unmixing algorithm. The Cuprite data set is a typical data set that is widely used for HSI classification and unmixing. This data set was captured by the AVIRIS sensor in June 1997, with 224 bands and a wavelength range of 0.4–2.5 $\mu$m, covering the Cuprite mining district in the state of Nevada. The original image was appropriately cropped to form a 250 $\times$ 191 pixel subimage for the experimentation. Based on the prior knowledge, bands 1–2, 105–115, 150–170, and 223–224 were removed due to their low SNR and strong water absorption effect. Finally, 188 bands were retained for the experimental analysis. The spectral library $\mathbf{A}\in\mathrm{R}^{188\times498}$ was built using 498 spectral signatures from the USGS library, for which low SNR and water absorption bands were also removed. Because a ground-truth abundance map is difficult to obtain, the mineral classification figure generated by Tricorder 3.3 software in 1995 [18] was used as a reference to quantitatively evaluate the performance of each method. The reference classification map is shown in Fig. 10, along with the results of the different algorithms. Three minerals, that is, alunite, buddingtonite, and chalcedony, are selected as representatives for the Cuprite mining district. The parameter settings for the different algorithms followed the recommendations in the literature [33], [37], to achieve the best performance, and parameters $\lambda_{\mathrm{TV}}$ and $\rho$ of the proposed SUSRLR-TV algorithm were set to 1e$-$3 and 1e$-$6, respectively.

In the estimated abundance maps shown in Fig. 10, all the algorithms can identify the locations of these three minerals, which demonstrates the effectiveness of the sparse unmixing method for real HSIs. However, some subtle differences in the estimated abundance maps can still be found. The main drawback of SUnSAL is that it obtains lower estimated abundance values in these mineral locations, compared with the other algorithms. Especially for chalcedony, SUnSAL has difficulty in distinguishing it from the background in the estimated abundance map. For the fractional abundances obtained by SUnSAL-TV and $MUA_{SLIC}$, over-smoothing is evident for both alunite and buddingtonite. The proposed SUSRLR-TV algorithm and the $S^2$WSU algorithm obtain results that are closer to the reference, especially for buddingtonite. However, the proposed algorithm shows a better performance for alunite, and the phenomenon of excessive smoothing is suppressed, to some degree.

Table III lists the computation time for each algorithm to perform unmixing on the Cuprite data. These run times were derived through the MATLAB R2017a platform using a personal computer with an Intel Core i3-8100 CPU at 3.60 GHz and 8-GB RAM. SUnSAL has the shortest execution time, as a result of the fact that it does not consider the spatial information. For the spatial regularizer method, $MUA_{SLIC}$, it shows good computing efficiency. Although there is no calculation of the $l_1$-norm, compared with SUnSAL-TV, the SVD operation adds extra time for the proposed SUSRLR-TV algorithm.

### *C. Analysis of the Parameter Sensitivity and Sparsity*

An analysis of the parameter sensitivity and sparsity was completed. In fact, the two regularization parameters $\lambda_{\mathrm{TV}}$ and $\rho$ do affect the performance of the proposed method, as they control the contributions of the two corresponding regularizations. TV regularization is used to promote the sparsity and further improve the smoothness of the abundance map [33], and the low-rank term is used to maintain the correlation of pixels within the homogeneous regions. To analyze the sensitivity of the parameters, the three simulated data sets at SNR $=$ 30 dB were used, and $\lambda_{\mathrm{TV}}$ and $\rho$ were set to [0, 5e$-$3, 1e$-$2, 1.5e$-$2, 2e$-$2, 2.5e$-$2, 3e$-$2, 3.5e$-$2, 4e$-$2, 4.5e$-$2, 5e$-$2]. The obtained SRE values for these two parameters are shown in Fig. 11.

From Fig. 11, it can be observed that parameter $\rho$ does not have a great effect on the model, and a satisfactory result can be obtained when $\rho$ is set within a certain range. However, if $\rho$ is set to 0, the SRE value drops sharply, which indicates that parameter $\rho$ does play a role in maintaining the local details of the abundance map. With regard to $\lambda_{\mathrm{TV}}$, the curve indicates that a large value causes the entire abundance map

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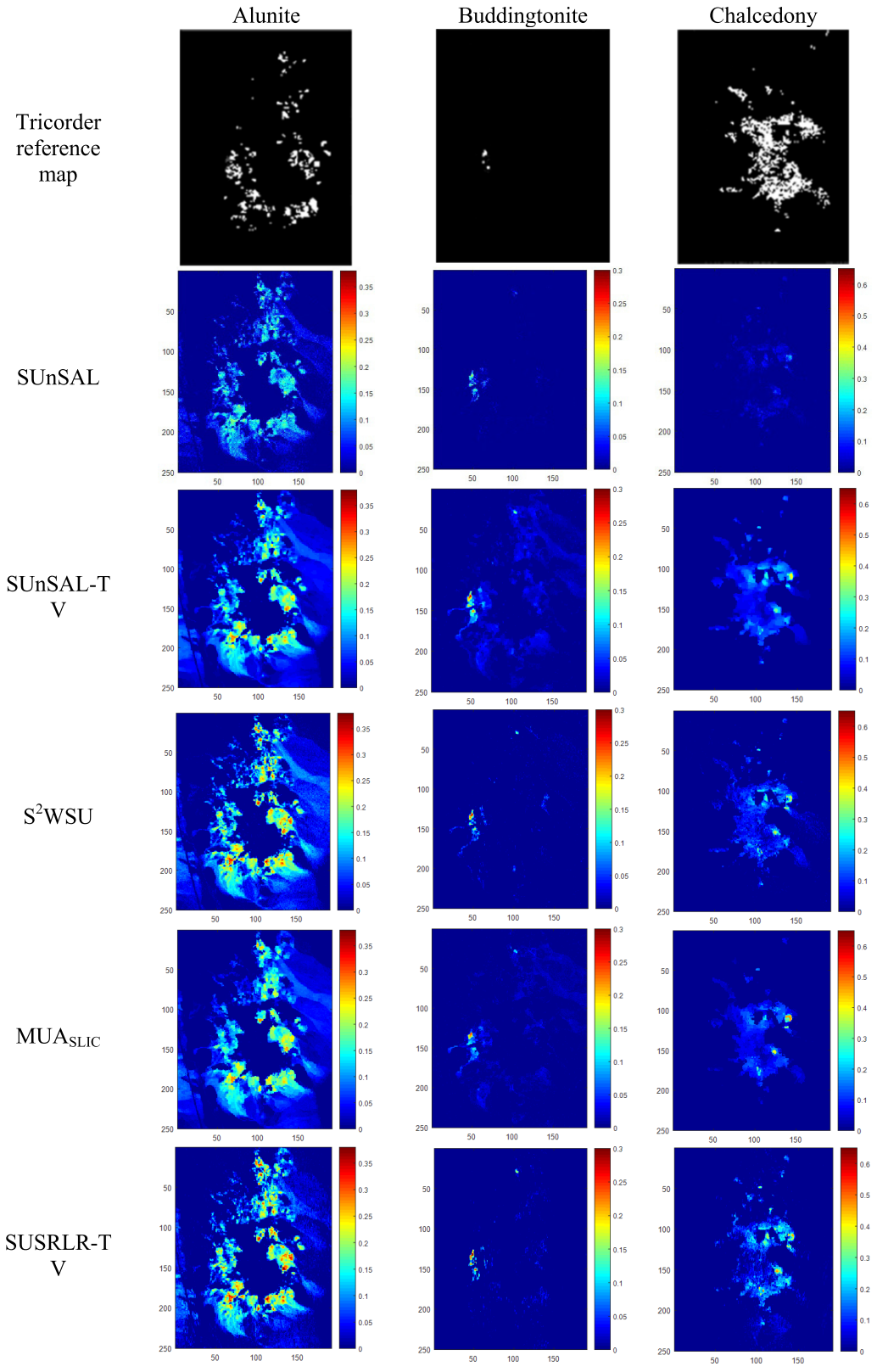



Fig. 10. Estimated abundance maps obtained by different algorithms for the Cuprite data set.

to be over-smoothed and eventually leads to a much smaller SRE. When the value of $\lambda_{\mathrm{TV}}$ is zero, the unmixing effect of the model is also good, as shown in Fig. 11(b) and (c). Based on these findings, it can be concluded that the parameters should be adjusted according to certain rules or the actual situation. For normalized hyperspectral data, we recommend that $\lambda_{\mathrm{TV}}$ is set from 1e$-$3 to 1e$-$2 under noise interference in a certain range (e.g., SNR $=$ 20 or 30 dB). Parameter $\rho$ is not

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TABLE III
AVERAGE PER-PIXEL PROCESSING TIMES OF EACH ALGORITHM ON THE CUPRITE DATA

| Algorithm | SUnSAL | SUnSAL-TV | $S^2$WSU | $MUA_{SLIC}$ | SUSRLR-TV |
|---|---|---|---|---|---|
| Time[s] | 0.00029 | 0.0292 | 0.0128 | 0.0025 | 0.0419 |

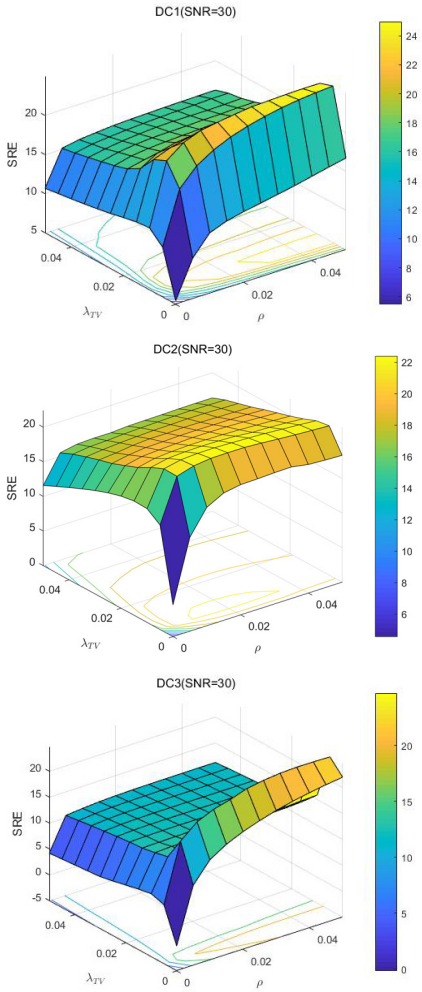



Fig. 11. SRE in relation to $\lambda_{\mathrm{TV}}$ and $\rho$ on different simulated data sets.

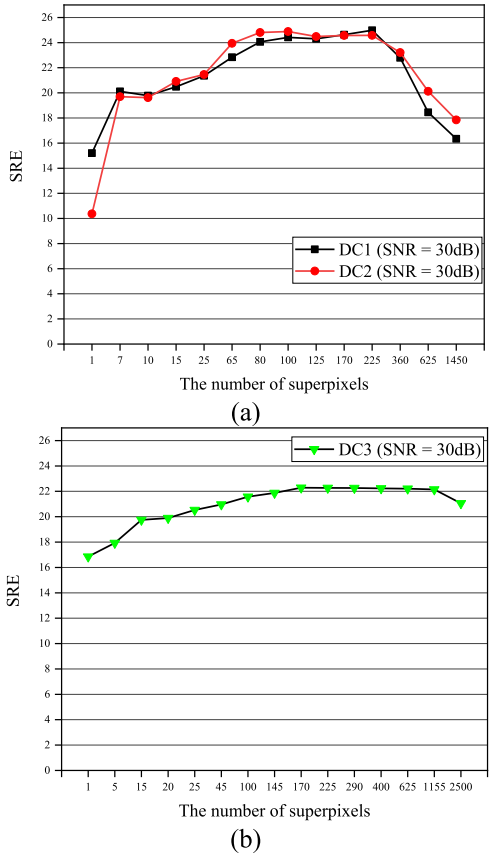



Fig. 12. SRE values at the number of superpixel. (a) DC1 and DC2. (b) DC3.

so sensitive, and the model can achieve a good performance when $\rho$ is set from 5e−3 to 1e−1.

To explore the effect of different superpixel segmentation scales on final unmixing, Fig. 12 shows the SRE values at a specific superpixel number. As in the previous analysis, it can be seen that a superpixel size that is too small or too large is not conducive to final unmixing. In addition, when the entire image is selected as a superpixel, the SRE value drops sharply, which verifies the local correlation assumption of the abundance matrix. In the simulation experiments undertaken in this study, each superpixel was made up of a certain number of pixels, in the range of tens to one hundred, which can lead to satisfactory results.

To verify the sparsity of the proposed method, the average abundance values of each endmember are plotted in Fig. 13. The ground truth for the three simulated data sets is also shown in the fourth column.

It can be found that even at an SNR of 10 dB, the estimated abundance matrices of DC1 and DC2 still show sparsity that is close to that of the ground truth. This reveals that the proposed SUSRLR-TV algorithm still exhibits sparsity, without the traditional $l_0$-norm or $l_1$-norm constraint.

## V. CONCLUSION

To obtain more reliable and accurate spatial information for spatial regularization sparse unmixing, the SUSRLR-TV sparse unmixing method has been proposed in this article. In the proposed method, the SLIC algorithm is used to obtain superpixels, with consideration of both the spectral and the spatial information of the hyperspectral imagery. In addition, to enhance the spatial continuity and suppress the effect of random noise, a reweighted low-rank constraint is imposed on the abundance matrix of each superpixel, and the local association of the abundance map is effectively maintained. Furthermore, to ensure the spatial relations between neighbor-

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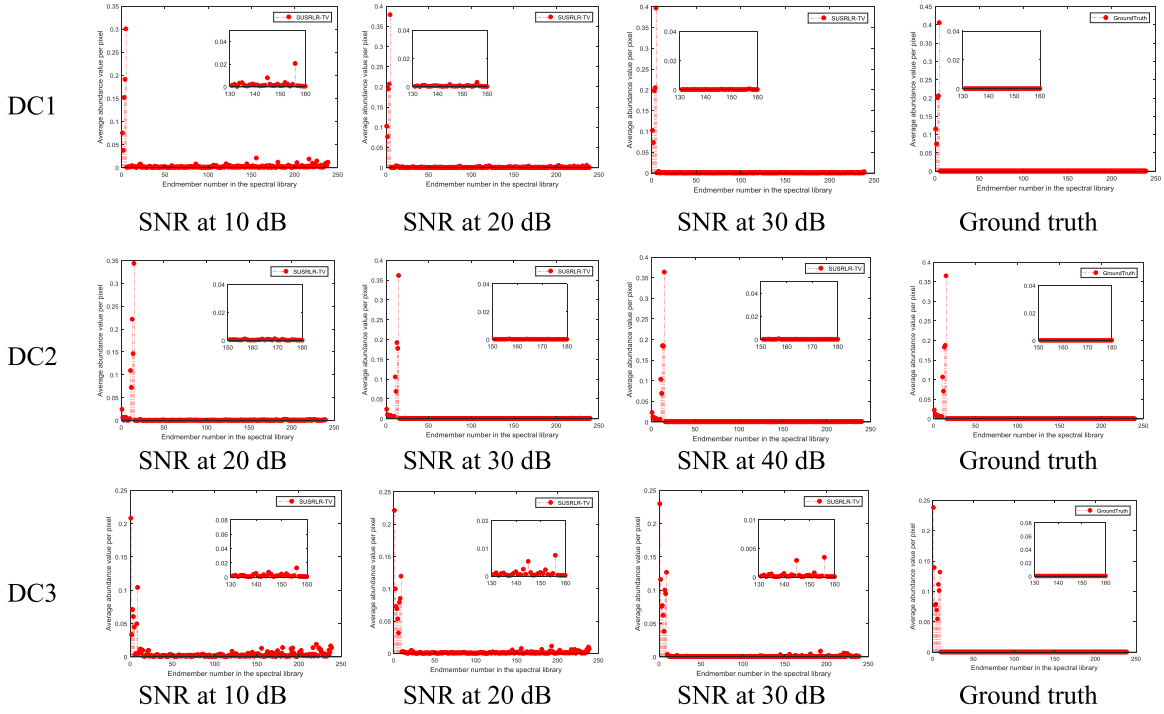



Fig. 13. Average abundance values of each endmember obtained by the proposed algorithm.

ing pixels, TV regularization is also enforced in the model, to promote smoothing of the abundance map and indirectly impose a sparse constraint. Finally, the ADMM is adopted for the objective function optimization.

To verify the effectiveness and superiority of the proposed algorithm, three sets of simulated data, together with the well-known Cuprite data set, were used to compare different sparse unmixing algorithms. The experimental results revealed that the proposed SUSRLR-TV algorithm outperforms the other algorithms both qualitatively and quantitatively, which further reveals that the use of efficient spatial information (considering both spectral similarity and spatial proximity) has a significant impact on the unmixing results. Although the current performance is very encouraging, the complexity of the proposed algorithm is still a challenge for practical applications, and a more efficient optimization algorithm will be studied in our future research.

## APPENDIX

The process of how to solve the SUSRLR-TV model is introduced in detail in this Appendix. To solve the proposed algorithm of (12), the ADMM strategy is adopted. We let $\mathbf{H}$ denote a linear operator computing the horizontal and the vertical differences between the components of $\mathbf{X}$ corresponding to the neighboring pixels for the TV model, and a clear objective function is rewritten as follows:

$$\min_{\mathbf{X}} \frac{1}{2}||\mathbf{Y}-\mathbf{AX}||_F^2+\lambda_{TV}||\mathbf{HX}||_{1,1} +\rho\sum_{i=1}^{S_k}\sum_{j=1}^{s}\left\|(\mathbf{X})_j^{(i)}\right\|_{w,*}+l_{R^+}(\mathbf{X}). \quad (14)$$

As the ADMM can split one complicated optimization problem into several suboptimization problems, auxiliary matrices $\mathbf{V}_1$, $\mathbf{V}_2$, $\mathbf{V}_3$, $\mathbf{V}_4$, $\mathbf{V}_5$, and $\mathbf{U}$ are introduced to make the objective function separable. These matrices should be added to augment the original function, and then the optimization problem defined in (14) can be reformulated as follows:

$$\begin{aligned}\min_{\mathbf{U},\mathbf{V}_1,\mathbf{V}_2,\mathbf{V}_3,\mathbf{V}_4,\mathbf{V}_5} &\frac{1}{2}||\mathbf{Y}-\mathbf{V}_1||_F^2+\lambda_{TV}||\mathbf{V}_3||_{1,1}\\ &+\rho\sum_{i=1}^{S_k}\sum_{j=1}^{s}\left\|(\mathbf{V}_4)_j^{(i)}\right\|_{w,*}+l_{R^+}(\mathbf{V}_5)\\ \text{s.t. }&\mathbf{V}_1=\mathbf{AU}\\ &\mathbf{V}_2=\mathbf{U}\\ &\mathbf{V}_3=\mathbf{HV}_2\\ &\mathbf{V}_4=\mathbf{U}\\ &\mathbf{V}_5=\mathbf{U}. \end{aligned} \quad (15)$$

The optimization (15) can then be written in a compact form as follows:

$$\min_{\mathbf{U},\mathbf{V}} \; g(\mathbf{V},\mathbf{U}) \text{ s.t. } \mathbf{GU}+\mathbf{BV}=\mathbf{0} \quad (16)$$

where

$$\begin{aligned}\mathbf{V}&\equiv(\mathbf{V}_1,\mathbf{V}_2,\mathbf{V}_3,\mathbf{V}_4,\mathbf{V}_5)\\ g(\mathbf{V},\mathbf{U})&=\frac{1}{2}||\mathbf{Y}-\mathbf{V}_1||_F^2+\lambda_{TV}||\mathbf{V}_3||_{1,1}\\ &\quad+\rho\sum_{i=1}^{S_k}\sum_{j=1}^{s}||(\mathbf{V}_4)_j^{(i)}||_{w,*}+l_{R^+}(\mathbf{V}_5).\end{aligned}$$

$$\mathbf{G}=\begin{bmatrix}\mathbf{A}\\ \mathbf{I}\\ \mathbf{0}\\ \mathbf{I}\\ \mathbf{I}\end{bmatrix}\quad \mathbf{B}=\begin{bmatrix}-\mathbf{I} & \mathbf{0} & \mathbf{0} & \mathbf{0} & \mathbf{0}\\ \mathbf{0} & -\mathbf{I} & \mathbf{0} & \mathbf{0} & \mathbf{0}\\ \mathbf{0} & \mathbf{H} & -\mathbf{I} & \mathbf{0} & \mathbf{0}\\ \mathbf{0} & \mathbf{0} & \mathbf{0} & -\mathbf{I} & \mathbf{0}\\ \mathbf{0} & \mathbf{0} & \mathbf{0} & \mathbf{0} & -\mathbf{I}\end{bmatrix} \quad (17)$$

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The augmented Lagrangian algorithm is used to solve problem (16), and turns it into an unconstrained problem, as shown in the following equation:

$$L(\mathbf{U},\mathbf{V},\mathbf{D}) = g(\mathbf{V},\mathbf{U}) + \frac{\mu}{2}||\mathbf{GU}-\mathbf{BV}-\mathbf{D}||_F^2 \quad (18)$$

where $\mu > 0$ is a penalty parameter. $\mathbf{D} = \{\mathbf{D}_1, \mathbf{D}_2, \mathbf{D}_3, \mathbf{D}_4, \mathbf{D}_5\}$ denotes the Lagrange multipliers associated with the constraint $\mathbf{GU}+\ \mathbf{BV}\ = 0$. The pseudocode to solve this optimization problem is listed in Algorithm 1. In practical implementations, the algorithm stops when the maximum iteration number is reached or when $||\mathbf{GU}^{(j)} + \mathbf{BV}^{(j)}||_F \le \sigma$, where $\sigma$ is a preset threshold.

The goal is to obtain the solution of each iteration of $\mathbf{U}$. By minimizing the formula containing the variable $\mathbf{U}$, the following optimization problem is obtained:

$$\begin{aligned}\mathbf{U}^{(k+1)} \leftarrow \arg\min_{\mathbf{U}} &\frac{\mu}{2}\left\|\mathbf{AU}-\mathbf{V}_1^{(k)}-\mathbf{D}_1^{(k)}\right\|_F^2 \\ &+\frac{\mu}{2}\left\|\mathbf{U}-\mathbf{V}_2^{(k)}-\mathbf{D}_2^{(k)}\right\|_F^2+\frac{\mu}{2}\left\|\mathbf{U}-\mathbf{V}_4^{(k)}-\mathbf{D}_4^{(k)}\right\|_F^2 \\ &+\frac{\mu}{2}\left\|\mathbf{U}-\mathbf{V}_5^{(k)}-\mathbf{D}_5^{(k)}\right\|_F^2. \end{aligned} \quad (19)$$

To find the partial derivative equation for $\mathbf{U}$ and make it equal to zero, the solution of $\mathbf{U}$ can be obtained as follows:

$$\mathbf{U}^{(k+1)} \leftarrow \left(\mathbf{A}^T\mathbf{A}+3\mathbf{I}\right)^{-1}\left(\mathbf{A}^T\xi_1+\xi_2+\xi_4+\xi_5\right) \quad (20)$$

where $\mathbf{I}$ is the identity matrix, $\mathbf{A}^T$ represents the transpose of $\mathbf{A}$, and $\xi_1 = \mathbf{V}_1^{(k)} + \mathbf{D}_1^{(k)}$, $\xi_2 = \mathbf{V}_2^{(k)} + \mathbf{D}_2^{(k)}$, $\xi_4 = \mathbf{V}_4^{(k)} + \mathbf{D}_4^{(k)}$, and $\xi_5 = \mathbf{V}_5^{(k)} + \mathbf{D}_5^{(k)}$.

The values of $\mathbf{V}_1$, $\mathbf{V}_2$, $\mathbf{V}_3$, $\mathbf{V}_4$, and $\mathbf{V}_5$ are then calculated in the current iteration. The solution of $\mathbf{V}_1$ is transformed into the following minimization problem:

$$\mathbf{V}_1^{(k+1)} \leftarrow \arg\min_{\mathbf{V}_1} \frac{1}{2}||\mathbf{Y}-\mathbf{V}_1||_F^2+\frac{\mu}{2}||\mathbf{AU}^{(k)}-\mathbf{V}_1-\mathbf{D}_1^{(k)}||_F^2. \quad (21)$$

$\mathbf{V}_1$ is obtained using the same strategy as $\mathbf{U}$, and the solution is

$$\mathbf{V}_1^{(k+1)} \leftarrow \frac{1}{1+\mu}\left[\mathbf{Y}+\mu\left(\mathbf{AU}^{(k)}-\mathbf{D}_1^{(k)}\right)\right]. \quad (22)$$

To obtain the solution of $\mathbf{V}_2$, the problem is translated into the following optimization problem:

$$\begin{aligned}\mathbf{V}_2^{(k+1)} \leftarrow \arg\min_{\mathbf{V}_2} &\frac{\mu}{2}\left\|\mathbf{U}^{(k)}-\mathbf{V}_2-\mathbf{D}_2^{(k)}\right\|_F^2 \\ &+\frac{\mu}{2}\left\|\mathbf{HV}_2-\mathbf{V}_3^{(k)}-\mathbf{D}_3^{(k)}\right\|_F^2. \end{aligned} \quad (23)$$

The same strategy can be used to obtain the following solution:

$$\mathbf{V}_2^{(k+1)} \leftarrow \mathbf{V}_2^{(k+1)} \leftarrow \left(\mathbf{H}^T\mathbf{H}+\mathbf{I}\right)^{-1}\left(\mathbf{U}^{(k)}-\mathbf{D}_2^{(k)}+\mathbf{H}^T\xi_3\right) \quad (24)$$

where $\xi_3 = \mathbf{V}_3^{(k)} + \mathbf{D}_3^{(k)}$.

To compute $\mathbf{V}_3$, we solve the optimization problem as follows:

$$\mathbf{V}_3^{(k+1)} \leftarrow \arg\min_{\mathbf{V}_3} \lambda_{TV}||\mathbf{V}_3||_{1,1}+\frac{\mu}{2}\left\|\mathbf{HV}_2^{(k)}-\mathbf{V}_3-\mathbf{D}_3^{(k)}\right\|_F^2 \quad (25)$$

whose solution is the well-known soft threshold [60]

$$\mathbf{V}_3^{(k+1)} \leftarrow \text{soft}\left(\mathbf{D}_3^{(k)}-\mathbf{HV}_2^{(k)}, \frac{\lambda_{TV}}{\mu}\right) \quad (26)$$

where $\text{soft}(\cdot,\tau)$ denotes the componentwise application of the soft-threshold function $\text{soft}(u,a) \equiv \text{sign}(u)\max\{|u|-a, 0\}$.

To compute $\mathbf{V}_4$, the optimization problem to be solved is

$$\begin{aligned}\mathbf{V}_4^{(k+1)} \leftarrow \arg\min_{\mathbf{V}_4} &\rho\sum_{i=1}^{S_k}\sum_{j=1}^{s}\left\|(\mathbf{V}_4)_j^{(i)}\right\|_{w,*} \\ &+\frac{\mu}{2}\left\|\mathbf{U}^{(k)}-\mathbf{V}_4-\mathbf{D}_4^{(k)}\right\|_F^2. \end{aligned} \quad (27)$$

For $(\mathbf{V}_4)_j^{(i)}$, the optimization problem is solved as follows:

$$\begin{aligned}\left(\mathbf{V}_4^{(k+1)}\right)_j^{(i)} \leftarrow \arg\min_{(\mathbf{V}_4)_j^{(i)}} &\rho\left\|(\mathbf{V}_4)_j^{(i)}\right\|_{w,*} \\ &+\frac{\mu}{2}\left\|\left(\mathbf{U}^{(k)}-\mathbf{V}_4-\mathbf{D}_4^{(k)}\right)_j^{(i)}\right\|_F^2 \end{aligned} \quad (28)$$

which is given by the weighted singular value thresholding ($w$-SVT) operator [57]

$$\left(\mathbf{V}_4^{(k+1)}\right)_j^{(i)} \leftarrow S_{w,\frac{\rho}{\mu}}\left(\left(\mathbf{U}^{(k)}-\mathbf{D}_4^{(k)}\right)_j^{(i)}\right) \quad (29)$$

where $S_{w,\tau}(\mathbf{X}) = \mathbf{U}(\Sigma-\tau\,\text{diag}(\mathbf{w}))_+\mathbf{V}^T$, where $\mathbf{U}\sum\mathbf{V}^{\mathrm{T}}$ is the SVD of $\mathbf{X}$ and $(x)_+ = \max\{x, 0\}$.

Finally, to compute $\mathbf{V}_5$, we solve the following optimization problem:

$$\mathbf{V}_5^{(k+1)} \leftarrow \arg\min_{\mathbf{V}_5} l_{R^+}(\mathbf{V}_5)+\frac{\mu}{2}\left\|\mathbf{U}^{(k)}-\mathbf{V}_5-\mathbf{D}_5^{(k)}\right\|_F^2. \quad (30)$$

In [61], the role of the $l_{\mathrm{R}+}$ term is to project the solution onto the nonnegative orthant, and the value of $\mathbf{V}_5$ is given by

$$\mathbf{V}_5^{(k+1)} \leftarrow \max\left(\mathbf{U}^{(k)}-\mathbf{D}_5^{(k)}, 0\right). \quad (31)$$

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## REFERENCES

[1] N. Huyan, X. Zhang, H. Zhou, and L. Jiao, "Hyperspectral anomaly detection via background and potential anomaly dictionaries construction," *IEEE Trans. Geosci. Remote Sens.*, vol. 57, no. 4, pp. 2263–2276, Apr. 2019.

[2] L. Zhuang, C. Lin, M. Figueiredo, and J. Bioucas-Dias, "Regularization parameter selection in minimum volume hyperspectral unmixing," *IEEE Trans. Geosci. Remote Sens.*, vol. 57, no. 12, pp. 9858–9877, Dec. 2019, doi: 10.1109/TGRS.2019.2929776.

[3] M. E. Paoletti *et al.*, "Capsule networks for hyperspectral image classification," *IEEE Trans. Geosci. Remote Sens.*, vol. 57, no. 4, pp. 2145–2160, Apr. 2019.

[4] J. Yao, D. Meng, Q. Zhao, W. Cao, and Z. Xu, "Nonconvex-sparsity and nonlocal-smoothness-based blind hyperspectral unmixing," *IEEE Trans. Image Process.*, vol. 28, no. 6, pp. 2991–3006, Jun. 2019.

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[5] Q. Wang, Z. Yuan, Q. Du, and X. Li, "GETNET: A general end-to-end 2-D CNN framework for hyperspectral image change detection," *IEEE Trans. Geosci. Remote Sens.*, vol. 57, no. 1, pp. 3–13, Jan. 2019.

[6] J. M. Bioucas-Dias *et al.*, "Hyperspectral unmixing overview: Geometrical, statistical, and sparse regression-based approaches," *IEEE J. Sel. Topics Appl. Earth Observ. Remote Sens.*, vol. 5, no. 2, pp. 354–379, Apr. 2012.

[7] J. M. Bioucas-Dias, A. Plaza, G. Camps-Valls, P. Scheunders, N. Nasrabadi, and J. Chanussot, "Hyperspectral remote sensing data analysis and future challenges," *IEEE Geosci. Remote Sens. Mag.*, vol. 1, no. 2, pp. 6–36, Jun. 2013.

[8] N. Keshava and J. F. Mustard, "Spectral unmixing," *IEEE Signal Process. Mag.*, vol. 19, no. 1, pp. 44–57, Jan. 2002.

[9] A. Plaza, Q. Du, J. M. Bioucas-Dias, X. Jia, and F. A. Kruse, "Foreword to the special issue on spectral unmixing of remotely sensed data," *IEEE Trans. Geosci. Remote Sens.*, vol. 49, no. 11, pp. 4103–4110, Nov. 2011.

[10] N. Dobigeon, J.-Y. Tourneret, C. Richard, J. C. M. Bermudez, S. McLaughlin, and A. O. Hero, "Nonlinear unmixing of hyperspectral images: Models and algorithms," *IEEE Signal Process. Mag.*, vol. 31, no. 1, pp. 82–94, Jan. 2014.

[11] R. Ammanouil, A. Ferrari, C. Richard, and S. Mathieu, "Nonlinear unmixing of hyperspectral data with vector-valued kernel functions," *IEEE Trans. Image Process.*, vol. 26, no. 1, pp. 340–354, Jan. 2017.

[12] B. Ayerdi and M. Graña, "Hyperspectral image nonlinear unmixing and reconstruction by ELM regression ensemble," *Neurocomputing*, vol. 174, pp. 299–309, Jan. 2016.

[13] A. Marinoni, J. Plaza, A. Plaza, and P. Gamba, "Nonlinear hyperspectral unmixing using nonlinearity order estimation and polytope decomposition," *IEEE J. Sel. Topics Appl. Earth Observ. Remote Sens.*, vol. 8, no. 6, pp. 2644–2654, Jun. 2015.

[14] R. Heylen, M. Parente, and P. Gader, "A review of nonlinear hyperspectral unmixing methods," *IEEE J. Sel. Topics Appl. Earth Observ. Remote Sens.*, vol. 7, no. 6, pp. 1844–1868, Jun. 2014.

[15] A. Halimi, P. Honeine, and J. M. Bioucas-Dias, "Hyperspectral unmixing in presence of endmember variability, nonlinearity, or mismodeling effects," *IEEE Trans. Image Process.*, vol. 25, no. 10, pp. 4565–4579, Oct. 2016.

[16] T. Imbiriba, J. C. M. Bermudez, and C. Richard, "Band selection for nonlinear unmixing of hyperspectral images as a maximal clique problem," *IEEE Trans. Image Process.*, vol. 26, no. 5, pp. 2179–2191, May 2017.

[17] J. J. Settle and N. A. Drake, "Linear mixing and the estimation of ground cover proportions," *Int. J. Remote Sens.*, vol. 14, no. 6, pp. 1159–1177, Apr. 1993.

[18] Z. Shi, T. Shi, M. Zhou, and X. Xu, "Collaborative sparse hyperspectral unmixing using $l_0$ norm," *IEEE Trans. Geosci. Remote Sens.*, vol. 56, no. 9, pp. 5495–5508, Sep. 2018.

[19] M. E. Winter, "N-FINDR: An algorithm for fast autonomous spectral end-member determination in hyperspectral data," *Proc. SPIE*, vol. 3753, pp. 266–275, Oct. 1999.

[20] J. M. P. Nascimento and J. M. B. Dias, "Vertex component analysis: A fast algorithm to unmix hyperspectral data," *IEEE Trans. Geosci. Remote Sens.*, vol. 43, no. 4, pp. 898–910, Apr. 2005.

[21] J. Boardman, F. A. Kruse, and R. O. Green, "Mapping target signatures via partial unmixing of AVIRIS data," in *Proc. JPL Airborne Earth Sci. Workshop*, vol. 1, 1995, pp. 23–26.

[22] T.-H. Chan, C.-Y. Chi, Y.-M. Huang, and W.-K. Ma, "A convex analysis-based minimum-volume enclosing simplex algorithm for hyperspectral unmixing," *IEEE Trans. Signal Process.*, vol. 57, no. 11, pp. 4418–4432, Nov. 2009.

[23] M. Arngren, M. N. Schmidt, and J. Larsen, "Unmixing of hyperspectral images using Bayesian non-negative matrix factorization with volume prior," *J. Signal Process. Syst.*, vol. 65, no. 3, pp. 479–496, Nov. 2011.

[24] W. He, H. Zhang, and L. Zhang, "Total variation regularized reweighted sparse nonnegative matrix factorization for hyperspectral unmixing," *IEEE Trans. Geosci. Remote Sens.*, vol. 55, no. 7, pp. 3909–3921, Jul. 2017.

[25] C. G. Tsinos, A. A. Rontogiannis, and K. Berberidis, "Distributed blind hyperspectral unmixing via joint sparsity and low-rank constrained non-negative matrix factorization," *IEEE Trans. Comput. Imag.*, vol. 3, no. 2, pp. 160–174, Jun. 2017.

[26] X. Wang, Y. Zhong, L. Zhang, and Y. Xu, "Spatial group sparsity regularized nonnegative matrix factorization for hyperspectral unmixing," *IEEE Trans. Geosci. Remote Sens.*, vol. 55, no. 11, pp. 6287–6304, Nov. 2017.

[27] X. Chen, J. Chen, X. Jia, B. Somers, J. Wu, and P. Coppin, "A quantitative analysis of virtual endmembers' increased impact on the collinearity effect in spectral unmixing," *IEEE Trans. Geosci. Remote Sens.*, vol. 49, no. 8, pp. 2945–2956, Aug. 2011.

[28] M.-D. Iordache, J. Bioucas-Dias, and A. Plaza, "Sparse unmixing of hyperspectral data," *IEEE Transactionss Geosci. Remote Sens.*, vol. 49, no. 6, pp. 2014–2039, Jun. 2011.

[29] J. M. Bioucas-Dias and M. A. T. Figueiredo, "Alternating direction algorithms for constrained sparse regression: Application to hyperspectral unmixing," in *Proc. 2nd Workshop Hyperspectral Image Signal Process., Evol. Remote Sens.*, Jun. 2010, pp. 1–4.

[30] J. Sigurdsson, M. O. Ulfarsson, and J. R. Sveinsson, "Blind hyperspectral unmixing using total variation and $\ell_q$ sparse regularization," *IEEE Trans. Geosci. Remote Sens.*, vol. 54, no. 11, pp. 6371–6384, Nov. 2016.

[31] X. Xu, Z. Shi, and B. Pan, "$\ell_0$-based sparse hyperspectral unmixing using spectral information and a multi-objectives formulation," *ISPRS J. Photogramm. Remote Sens.*, vol. 141, pp. 46–58, Jul. 2018.

[32] M.-D. Iordache, J. M. Bioucas-Dias, and A. Plaza, "Collaborative sparse regression for hyperspectral unmixing," *IEEE Trans. Geosci. Remote Sens.*, vol. 52, no. 1, pp. 341–354, Jan. 2014.

[33] M.-D. Iordache, J. M. Bioucas-Dias, and A. Plaza, "Total variation spatial regularization for sparse hyperspectral unmixing," *IEEE Trans. Geosci. Remote Sens.*, vol. 50, no. 11, pp. 4484–4502, Nov. 2012.

[34] S. Zhang, J. Li, H.-C. Li, C. Deng, and A. Plaza, "Spectral–spatial weighted sparse regression for hyperspectral image unmixing," *IEEE Trans. Geosci. Remote Sens.*, vol. 56, no. 6, pp. 3265–3276, Jun. 2018.

[35] J. Li, Y. Li, R. Song, S. Mei, and Q. Du, "Local spectral similarity preserving regularized robust sparse hyperspectral unmixing," *IEEE Trans. Geosci. Remote Sens.*, vol. 57, no. 10, pp. 7756–7769, Oct. 2019.

[36] Z. Li, J. Chen, and S. Rahardja, "Superpixel construction for hyperspectral unmixing," in *Proc. 26th Eur. Signal Process. Conf. (EUSIPCO)*, Rome, Italy, Sep. 2018, pp. 647–651.

[37] R. A. Borsoi, T. Imbiriba, J. C. M. Bermudez, and C. Richard, "A fast multiscale spatial regularization for sparse hyperspectral unmixing," *IEEE Geosci. Remote Sens. Lett.*, vol. 16, no. 4, pp. 598–602, Apr. 2019.

[38] Y. Chen, W. He, N. Yokoya, and T.-Z. Huang, "Blind cloud and cloud shadow removal of multitemporal images based on total variation regularized low-rank sparsity decomposition," *ISPRS J. Photogramm. Remote Sens.*, vol. 157, pp. 93–107, Nov. 2019.

[39] W. He, N. Yokoya, L. Yuan, and Q. Zhao, "Remote sensing image reconstruction using tensor ring completion and total variation," *IEEE Trans. Geosci. Remote Sens.*, vol. 57, no. 11, pp. 8998–9009, Nov. 2019.

[40] Y. Wang, X. Chen, Z. Han, and S. He, "Hyperspectral image super-resolution via nonlocal low-rank tensor approximation and total variation regularization," *Remote Sens.*, vol. 9, no. 12, p. 1286, 2017.

[41] Q. Wang, Z. Wu, J. Jin, T. Wang, and Y. Shen, "Low rank constraint and spatial spectral total variation for hyperspectral image mixed denoising," *Signal Process.*, vol. 142, pp. 11–26, Jan. 2018.

[42] S. Li, W. Wang, H. Qi, B. Ayhan, C. Kwan, and S. Vance, "Low-rank tensor decomposition based anomaly detection for hyperspectral imagery," in *Proc. IEEE Int. Conf. Image Process. (ICIP)*, Sep. 2015, pp. 4525–4529.

[43] M. A. Veganzones, J. E. Cohen, R. Cabral Farias, J. Chanussot, and P. Comon, "Nonnegative tensor CP decomposition of hyperspectral data," *IEEE Trans. Geosci. Remote Sens.*, vol. 54, no. 5, pp. 2577–2588, May 2016.

[44] Y. Qian, F. Xiong, S. Zeng, J. Zhou, and Y. Y. Tang, "Matrix-vector nonnegative tensor factorization for blind unmixing of hyperspectral imagery," *IEEE Trans. Geosci. Remote Sens.*, vol. 55, no. 3, pp. 1776–1792, Mar. 2017.

[45] T. Imbiriba, R. A. Borsoi, and J. C. M. Bermudez, "A low-rank tensor regularization strategy for hyperspectral unmixing," in *Proc. IEEE Stat. Signal Process. Workshop (SSP)*, Jun. 2018, pp. 373–377.

[46] T. Imbiriba, R. A. Borsoi, and J. C. M. Bermudez, "Low-rank tensor modeling for hyperspectral unmixing accounting for spectral variability," *IEEE Trans. Geosci. Remote Sens.*, vol. 58, no. 3, pp. 1833–1842, Mar. 2020, doi: 10.1109/TGRS.2019.2949543.

[47] F. Xiong, Y. Qian, J. Zhou, and Y. Y. Tang, "Hyperspectral unmixing via total variation regularized nonnegative tensor factorization," *IEEE Trans. Geosci. Remote Sens.*, vol. 57, no. 4, pp. 2341–2357, Apr. 2019.

[48] P. V. Giampouras, K. E. Themelis, A. A. Rontogiannis, and K. D. Koutroumbas, "Simultaneously sparse and low-rank abundance matrix estimation for hyperspectral image unmixing," *IEEE Trans. Geosci. Remote Sens.*, vol. 54, no. 8, pp. 4775–4789, Aug. 2016.

[49] S. S. Chen, D. L. Donoho, and M. A. Saunders, "Atomic decomposition by basis pursuit," *SIAM Rev.*, vol. 43, no. 1, pp. 129–159, Jan. 2001.

[50] E. J. Candès, J. Romberg, and T. Tao, "Stable signal recovery from incomplete and inaccurate measurements," *Commun. Pure Appl. Math.*, vol. 59, no. 8, pp. 1207–1223, Aug. 2006.

[51] E. J. Candes and T. Tao, "Decoding by linear programming," *IEEE Trans. Inf. Theory*, vol. 51, no. 12, pp. 4203–4215, Dec. 2005.


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[52] R. Achanta, A. Shaji, K. Smith, A. Lucchi, P. Fua, and S. Süsstrunk, "SLIC superpixels compared to state-of-the-art superpixel methods," *IEEE Trans. Pattern Anal. Mach. Intell.*, vol. 34, no. 11, pp. 2274–2282, Nov. 2012.
[53] X. Xu, J. Li, C. Wu, and A. Plaza, "Regional clustering-based spatial preprocessing for hyperspectral unmixing," *Remote Sens. Environ.*, vol. 204, pp. 333–346, Jan. 2018.
[54] Ren and Malik, "Learning a classification model for segmentation," in *Proc. 9th IEEE Int. Conf. Comput. Vis.*, Oct. 2003, pp. 10–17.
[55] J.-F. Cai, E. J. Candès, and Z. Shen, "A singular value thresholding algorithm for matrix completion," *SIAM J. Optim.*, vol. 20, no. 4, pp. 1956–1982, Jan. 2010.
[56] E. J. Candés, M. B. Wakin, and S. P. Boyd, "Enhancing sparsity by reweighted $\ell_1$ minimization," *J. Fourier Anal. Appl.*, vol. 14, nos. 5–6, pp. 877–905, Dec. 2008.
[57] W. Dong, G. Shi, X. Li, Y. Ma, and F. Huang, "Compressive sensing via nonlocal low-rank regularization," *IEEE Trans. Image Process.*, vol. 23, no. 8, pp. 3618–3632, Aug. 2014.
[58] L. Tong, J. Zhou, X. Li, Y. Qian, and Y. Gao, "Region-based structure preserving nonnegative matrix factorization for hyperspectral unmixing," *IEEE J. Sel. Topics Appl. Earth Observ. Remote Sens.*, vol. 10, no. 4, pp. 1575–1588, Apr. 2017.
[59] R. Ammanouil, A. Ferrari, and C. Richard, "A graph Laplacian regularization for hyperspectral data unmixing," in *Proc. IEEE Int. Conf. Acoust., Speech Signal Process. (ICASSP)*, Apr. 2015, pp. 1637–1641.
[60] P. L. Combettes and V. R. Wajs, "Signal recovery by proximal forward-backward splitting," *Multiscale Model. Simul.*, vol. 4, no. 4, pp. 1168–1200, Jan. 2005.
[61] A. Zare and P. Gader, "Hyperspectral band selection and endmember detection using sparsity promoting priors," *IEEE Geosci. Remote Sens. Lett.*, vol. 5, no. 2, pp. 256–260, Apr. 2008.



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